

Higher Normal School of Technical Education of Mohammedia
Department: Electrical Engineering

Final Year Project Thesis
Presented for the award of the degree of
State Engineer in Electrical Engineering and Renewable Energy

**Development of an Embedded System for
Intelligent Energy Management and
Monitoring between Two Interconnected
Renewable Microgrids**

Defended on November 8, 2025

Defended by:

Mr. BOUKHACHA ELIDRISSI
Naoufal

Committee Members:

Pr. ELKHAILI Mohamed
ENSET-M - President

Mr. EL ARFAOUI Adil
Examiner

Pr. M. BAKHOUYA
UIR - Supervisor

Mr. ELAMRANI Abderrahmane
UIR - Adviser

Academic Year: 2025-2026

Dedication

*

Firstly, I dedicate this work to my late uncle, who was more than an uncle to me — he was a father, a friend, and a guiding presence in my life.

I also dedicate this work to my beloved parents and family, whose love and support have always been my greatest strength. To all the teachers who guided and inspired me throughout my academic journey. And to my dear friends, for their encouragement and companionship along the way.

◇

— *Naoufal*

Acknowledgements

First and foremost, I thank God, the Almighty and Most Benevolent, who has given me the strength, courage, and patience to complete this work.

I extend my heartfelt thanks to my supervisor, **Prof. M. Bakhouya** from the International University of Rabat, for his trust, availability, and especially his insightful guidance and support throughout the entire duration of this project. His advice significantly enriched my reflections and ensured this endeavor's success.

I also express my warmest thanks to the jury members for their interest in my research. Their willingness to examine my work and enhance it with their valuable suggestions is deeply appreciated.

My sincere gratitude goes to the entire teaching staff of the **Engineering Cycle in Electrical Engineering and Renewable Energies** at **ENSET Mohammedia**. Their support and the opportunities they provided have been instrumental throughout my years of study.

Special thanks to the members of the **EEBLAB team** for their collaboration and support. It was a great privilege to work together within the framework of their research activities.

Lastly, I am profoundly grateful to my family for their prayers, encouragement, and unconditional support, both morally and spiritually. Their unwavering belief in me has been a vital force in realizing this work.

Contents

Acknowledgements	i
List of Figures	viii
List of Tables	ix
List of Acronyms	x
Abstract	xi
General Introduction	1
1 Host Organization - University UIR and LERMA Laboratory	3
Introduction	3
1.1 University UIR	3
1.1.1 History and Mission	3
1.1.2 Academic and Research Excellence	4
1.1.3 Contributions to Science and Technology	4
1.2 LERMA Laboratory	5
1.2.1 Role and Significance	5
1.2.2 Key Projects and Achievements	5
1.2.3 Research Themes and Expertise	5
1.3 The Energy Efficient Building (EEB) Platform	6
1.3.1 Purpose and Strategic Importance	6
1.3.2 Contributions to Sustainable Solutions	6
1.3.3 Technologies and Methodologies	7
1.4 Impact and Future Prospects	7
1.4.1 Influence on Research Landscape	7
1.4.2 Future Directions and Projects	7
Conclusion	8

2	State of the Art	9
	Introduction	9
2.1	Background and Context	9
2.1.1	Microgrids: Definition and Importance	9
2.1.2	Renewable Energy Sources in Microgrids	10
2.1.2.1	Photovoltaic Systems	10
2.1.2.2	Wind Turbines	12
2.1.3	Energy Storage Systems	14
2.1.3.1	Battery Technologies	14
2.1.3.2	Battery State of Charge State of Charge (SOC)	14
2.1.4	SOC Estimation Methods	15
2.1.4.1	Voltage-Based Estimation	15
2.1.4.2	Coulomb Counting	15
2.1.4.3	Model-Based and AI Approaches	16
2.1.4.4	Comparison and Limitations	17
2.1.5	Embedded Systems for Energy Monitoring	17
2.1.6	Supervisory Control and Data Acquisition (SCADA) for Microgrids	18
2.1.7	Switching and Load Management in Microgrids	18
2.2	Literature Review	19
2.2.1	Renewable Energy Sources in Microgrids	19
2.2.2	Energy Storage and SOC-Based Management	19
2.2.3	Embedded Systems for Data Acquisition	20
2.2.4	Data Processing and Local Control Units	21
2.2.5	Supervisory Control and Communication in Microgrids	21
2.2.6	Switching and Load Management Strategies	22
2.2.7	Summary and Research Gap	22
2.3	Problem Statement and Objectives	23
2.3.1	Problem Statement	23
2.3.2	Motivation	24
2.3.3	Project Objectives	25
	Conclusion	26
3	System Design	27
	Introduction	27
3.1	Overall System Architecture	27
3.1.1	Concept of Dual Microgrids	27
3.1.1.1	Description of Microgrid 1 (MG1)	27
3.1.1.2	Description of Microgrid 2 (MG2)	30
3.1.2	Block Diagram of the Proposed System	33

3.1.3	Functional Layers of the System	34
3.2	Hardware Design	35
3.2.1	Measurement Circuits	35
3.2.1.1	PCB1: PV2 Measurement	36
3.2.1.2	PCB2: Battery Monitoring	38
3.2.1.3	Data Acquisition from MG1 Sources	39
3.2.2	Switching Circuit (PCB3)	40
3.3	Software Design	41
3.3.1	Embedded Software (Arduino)	41
3.3.2	Raspberry Pi Scripts (Python)	42
3.3.3	Node-RED Flows	42
3.3.4	ThingsBoard Supervision	43
3.4	Communication Architecture	43
3.5	Control and Switching Logic	46
3.5.1	Operation Modes	46
3.5.2	SOC-Based Switching Algorithm	46
3.5.3	Flowchart of Decision Logic	47
3.6	System Diagrams and Illustrations	47
	Conclusion	48
4	Implementation and Validation	49
	Introduction	49
4.1	Hardware Implementation	49
4.2	Hardware Implementation	49
4.2.1	PCB-1: PV2 Measurement Board	50
4.2.2	PCB-2: Battery Monitoring Board	50
4.2.3	PCB-3: Switching Module	51
4.2.4	Integration of External Measurements	52
4.2.5	Raspberry Pi Central Unit	53
4.3	Software Implementation	53
4.3.1	Data Acquisition Layer	54
4.3.2	Python Processing Layer	55
4.3.3	Communication Layer	56
4.3.4	SOC-Based Switching Algorithm	56
4.3.5	Processed Results	57
4.4	Dashboard and Supervision (ThingsBoard)	58
4.4.1	Data Transmission and Command Flow	58
4.4.2	Dashboard Structure	59
4.4.3	ThingsBoard Device and Alias Configuration	60

4.4.4	Access and User Roles	61
4.4.5	Supervision Windows and Interface Snapshots	61
4.5	Experimental Validation	63
4.5.1	Global System Setup	63
4.5.2	Validation of PV Production	63
4.5.3	Validation of Wind Turbine Production	64
4.5.4	Validation of Interconnection and SOC-Based Switching	65
4.5.5	Discussion of Results	65
	Conclusion	66
	General Conclusion	67
	A Dashboard Windows (ThingsBoard)	69
A.1	Home Dashboard	69
A.2	Global Dashboard	70
A.3	Battery Dashboard	70
A.4	Microgrid 1 Dashboard	71
A.5	Microgrid 2 Dashboard	71
A.6	Modes and Control Dashboard	72
A.7	Flow and Transfer Dashboard	72
A.8	History Dashboard	73
	B Validation Results	74
B.1	PV Validation	74
B.1.1	PV ₁ Energy Validation	74
B.1.2	PV ₂ Energy Validation	75
B.1.3	PV ₁ Power Validation	75
B.1.4	PV ₂ Power Validation	76
B.2	Wind Validation	76
B.2.1	Wind Power Validation	76
B.3	Switch-Based Control Validation	77
B.3.1	Battery and Load Management	77

List of Figures

1.1	Laboratories of UIR	4
1.2	LERMA Laboratory	5
1.3	LERMA Projects	5
1.4	Energy Efficient Building (EEB) Platform	6
1.5	Key Projects under EEBLAB	7
2.1	Schematic of a microgrid [1]	10
2.2	Comparison between PWM and MPPT charge controllers [2].	12
2.3	Comparison between HAWT and VAWT designs.	13
2.4	Conventional isolated microgrid without SOC-based management.	24
3.1	Silent Wind 12V wind turbine [37]	28
3.2	SL50CE-18 Solar Panel [38]	29
3.3	Silent Wind 12V Hybrid Charge Controller [39]	29
3.4	OUTDO 12V/150Ah Gel Battery [40]	30
3.5	Load 1: CO ₂ -controlled ventilation system [41]	30
3.6	Amerisolar AS-6P30-265W photovoltaic panel [42]	31
3.7	Victron BlueSolar MPPT 150/35 Charge Controller [43])	32
3.8	Victron Energy 12V/130Ah GEL Battery [44]	33
3.9	Load 2: Atlantic 12V Portable Refrigerator	33
3.10	General architecture of MG1 and MG2 with interconnection link.	34
3.11	Triangular representation of the functional layers	35
3.12	Arduino Nano used as the main acquisition unit [45]	36
3.13	ACS712 current sensor module	36
3.14	PCB1 Architecture for PV2 measurement.	37
3.15	PCB1 schematic for PV2 measurement	37
3.16	PCB2 Arcitecture for monitoring Battery 1 and Battery 2.	38
3.17	PCB2 schematic for monitoring Battery 1 and Battery 2.	39
3.18	Bluetooth-based data acquisition from MG1 sources	39
3.19	PCB3 Arcitecture for switching circuit	40
3.20	PCB3 schematic for switching circuit	40
3.21	Communication architecture – Data acquisition layer	45

3.22	Communication architecture – Data processing and supervision layer	45
3.23	Flowchart of SOC-based switching algorithm with hysteresis.	47
3.24	Global architecture of the proposed system, showing energy sources, bat- teries, monitoring boards, and switching module.	48
4.1	Prototype of PCB-1	50
4.2	Prototype of PCB-2.	51
4.3	Prototype of PCB-3.	51
4.4	Weather station and external sensors	52
4.5	Raspberry Pi central unit connected to PCBs and Bluetooth receiver for data aggregation and processing.	53
4.6	Node-RED flow for data acquisition from PCB-1 (PV2 board).	54
4.7	Node-RED flow for data acquisition from PCB-2 (Baterries board).	54
4.8	Example of CSV log file generated by Node-RED	54
4.9	Node-RED flow for sending telemetry data to ThingsBoard via HTTP. . .	58
4.10	Node-RED flow for receiving and parsing RPC commands from ThingsBoard. .	59
4.11	Node-RED flow to sync ThingsBoard attributes with real system state. . .	59
4.12	Device configuration in ThingsBoard for microgrid supervision.	60
4.13	Entity alias configuration linking telemetry to dashboard widgets.	61
4.14	Global dashboard: microgrids overview, battery SOC, daily production, load consumption.	61
4.15	Battery dashboard: SOC, voltage, net power, energy, charge/discharge. . .	62
4.16	Modes & control dashboard: current/recommended mode, auto/manual switch, manual controls.. . . .	62
4.17	Global experimental setup	63
4.18	Validation of cumulative energy production of PV_1 with respect to solar irradiation.	64
4.19	Qualitative validation of wind turbine production: power versus wind speed. .	64
4.20	Experimental validation of interconnection and SOC-based switching be- tween the two microgrids.	65
A.1	Home dashboard: general description of the project and navigation button to access other states.	69
A.2	Global dashboard: overview of both microgrids, SOC of Bat1 and Bat2, daily energy production, load consumptions, and total balance.	70
A.3	Battery dashboard: SOC, voltage, net current, net power, cumulative en- ergy, and charge/discharge states.	70
A.4	Microgrid 1 dashboard: PV_1 production, wind turbine generation, load 1 consumption, and cumulative energies with daily balance.	71

A.5	Microgrid 2 dashboard: PV2 production, load 2 consumption, shared energy with MG1, and daily balance.	71
A.6	Modes & control dashboard: recommended mode, confirmed mode, auto/-manual selector, and manual control buttons.	72
A.7	Flow and transfer dashboard: cumulative energy shared between microgrids and timeline of switching modes.	72
A.8	History dashboard: table of all parameters with selectable time range. . . .	73
B.1	Validation of cumulative energy production of PV_1 with respect to solar irradiation.	74
B.2	Validation of cumulative energy production of PV_2 with respect to solar irradiation.	75
B.3	Validation of instantaneous power of PV_1 with respect to solar irradiation.	75
B.4	Validation of instantaneous power of PV_2 with respect to solar irradiation.	76
B.5	Validation of wind turbine power production with respect to wind speed. .	76
B.6	Validation of switch-based control strategy for battery and load management.	77

List of Tables

3.1	Specifications of Silent Wind 12V Turbine [36]	28
3.2	Specifications of SL50CE-18 Solar Panel [38]	29

List of Acronyms

AI	Artificial Intelligence
ATS	Automatic Transfer Switch
BMS	Battery Management Systems
Batt1	Battery 1
Batt2	Battery 2
DAQ	Data Acquisition
DERs	Distributed Energy Resources
ECM	Equivalent Circuit Model
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
Load1	Load of Microgrid 1
Load2	Load of Microgrid 2
MG	Microgrid
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracking
MQTT	Message Queuing Telemetry Transport
OCV	Open-Circuit Voltage
PCB	Printed Circuit Boards
PV	Photovoltaic
PWM	Pulse-Width Modulation
RES	Renewable Energy Sources
SCADA	Supervisory Control and Data Acquisition
SOC	State of Charge
WECS	Wind Energy Conversion System
WT	Wind Turbines

Abstract

This internship project focuses on the design and implementation of an intelligent energy management system for interconnected microgrids powered by renewable energy sources. The system integrates photovoltaic panels, a wind turbine, and two battery storage units, with automatic switching based on the SOC of the batteries.

To achieve this, three custom Printed Circuit Boards (PCB) were developed: the first to measure the voltage and current of a photovoltaic source, the second to monitor charging and discharging currents of the two batteries, and the third to perform switching operations through a relay-based architecture controlled by an Arduino. A Raspberry Pi was employed as the central processing unit, collecting real-time data from the sensors and additional sources via Bluetooth, while performing SOC and energy balance calculations through Python scripts.

Node-RED was used as a middleware to manage communications between the Raspberry Pi, the cloud supervision platform ThingsBoard (via HTTP and MQTT), and the Arduino-based switching system. The developed system successfully demonstrated automatic and reliable switching between batteries and microgrids, while enabling supervision, data analysis, and performance evaluation of renewable energy production under varying environmental conditions.

General Introduction

The rapid growth in global energy demand, combined with environmental concerns and the progressive depletion of fossil fuel reserves, has led to an accelerated shift toward renewable energy sources (RES).

Among these, solar and wind energy, along with other distributed resources, have emerged as viable alternatives to conventional generation. They offer cleaner and more sustainable solutions, contributing to global climate objectives and sustainable development goals. Nevertheless, the inherent intermittency and variability of renewable energy present critical challenges, particularly regarding system stability, reliability, and efficient energy management.

In this context, microgrids have gained increasing attention as decentralized energy systems capable of integrating various renewable resources with advanced energy storage solutions and intelligent load management strategies.

They offer significant benefits in terms of flexibility, resilience, and autonomy, while also playing a strategic role in the broader energy transition. To achieve these objectives, it becomes essential to develop robust monitoring platforms, accurate State of Charge (SOC) estimation methods, efficient embedded acquisition systems, and advanced supervisory control architectures. Together, these components ensure optimal operation, improved reliability, and sustainable performance of microgrids.

This report addresses these aspects through the design, implementation, and evaluation of a complete microgrid monitoring and control platform. The work encompasses multiple dimensions, including:

- the modeling and integration of renewable energy sources,
- the estimation and supervision of battery State of Charge,
- the development of embedded systems for acquisition and real-time data processing,
- the implementation of supervisory control layers based on modern SCADA and IoT technologies.

Special emphasis is placed on intelligent switching strategies and advanced load management mechanisms, aiming to ensure both operational continuity and overall system efficiency.

The document is structured as follows:

- **Chapter 1:** introduces the host organization and presents its research environment.
- **Chapter 2:** provides a detailed state of the art, covering microgrid architectures, renewable integration, SOC estimation approaches, embedded acquisition platforms, and supervisory control strategies.
- **Chapter 3:** focuses on the design and hardware implementation of the proposed system, followed by the development of the required software layers.
- **Chapter 4:** presents the experimental validation, results, and critical analysis.
- **Conclusion:** summarizes the main contributions of this work, highlights its limitations, and suggests perspectives for future improvements and research directions.

Chapter 1

Host Organization - University UIR and LERMA Laboratory

Introduction

The International University of Rabat (UIR) is a prominent educational institution in Morocco, dedicated to merging tradition with modernity and fostering an environment of innovation and intellectual advancement. Established in 2009, UIR has made significant strides in higher education and research, positioning itself as a leader in scientific inquiry and technological development. Central to UIR's mission is the commitment to providing cutting-edge education and promoting groundbreaking research, aimed at shaping future leaders who will contribute significantly to Morocco's intellectual and social landscape.

This chapter delves into the foundational aspects of UIR, its academic and research excellence, and its substantial contributions to science and technology. It also highlights the pivotal role of the LERMA Laboratory within UIR's research infrastructure, focusing on its research themes, expertise, key projects, and achievements. Furthermore, the chapter explores the strategic importance of the Energy Efficient Building (EEB) Platform, its innovative technologies and methodologies, and its contributions to sustainable development. Finally, the chapter reflects on the impact of UIR on the research landscape and outlines future prospects and projects that underscore UIR's commitment to excellence and innovation.

1.1 University UIR

1.1.1 History and Mission

The International University of Rabat (UIR) stands as a leading educational institution in Morocco, envisioned by Professor Nouredine Mouaddib in 2006. Officially established in 2009, UIR signed a development contract with the Moroccan government, affirming its

dedication to academic excellence. The foundation of UIR was symbolically marked in 2010 by King Mohammed VI, who laid the first stone of the campus, thus initiating a journey towards innovation and intellectual advancement.

UIR's mission is deeply rooted in delivering cutting-edge higher education and promoting groundbreaking research. The university strives to merge tradition with modernity, shaping future leaders who will play significant roles in Morocco's intellectual and social spheres. UIR's vision highlights the importance of adhering to global educational standards while maintaining a strong connection to local cultural and social contexts.

1.1.2 Academic and Research Excellence

UIR is acclaimed for its academic rigor and research excellence. The university provides a diverse range of academic programs tailored to meet the dynamic demands of various industries. These programs are supported by cutting-edge research initiatives that make substantial contributions to science and technology. UIR is home to multiple centers of excellence, each dedicated to advancing knowledge and promoting innovation in specialized fields. These centers play a crucial role in fostering an environment of intellectual growth and discovery, reinforcing UIR's status as a hub of academic and research excellence.



Figure 1.1: Laboratories of UIR

1.1.3 Contributions to Science and Technology

The contributions of UIR to science and technology are substantial. Through its diverse research programs and collaborations, the university has positioned itself as a leader in scientific inquiry and technological advancement. UIR's research outputs are not only published in renowned journals but also have practical applications that benefit society.

1.2 LERMA Laboratory

1.2.1 Role and Significance

The LERMA Laboratory is a cornerstone of UIR's research infrastructure. It plays a crucial role in enhancing the university's research capabilities and its academic prestige. LERMA's significance extends beyond UIR, contributing to the broader scientific community through its innovative research and development efforts.



Figure 1.2: LERMA Laboratory

1.2.2 Key Projects and Achievements

LERMA has undertaken numerous high-impact projects that have garnered significant recognition. These projects range from developing new renewable energy technologies to creating sustainable solutions for urban development. The laboratory's achievements are a testament to its commitment to excellence and its ability to translate research into tangible benefits.

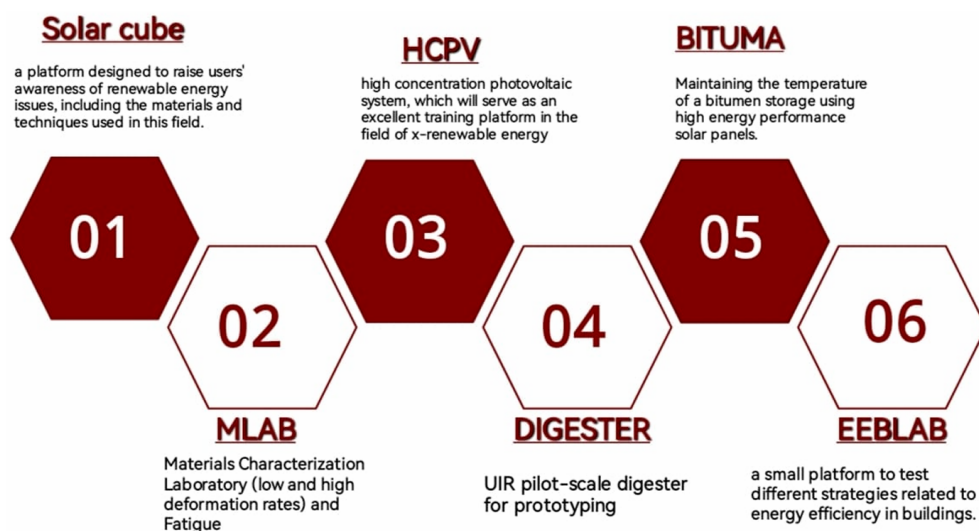


Figure 1.3: LERMA Projects

1.2.3 Research Themes and Expertise

LERMA's research activities are oriented towards the design and development of smart materials, structures, and systems used in aeronautical, automotive, and energetic tech-

nologies. It mainly hosts the research activities conducted by faculties and research scientists from four engineering schools: AERO, AUTO, ESIE, and ESIN. Researchers and scientists focus on developing methods and tools required to design large-scale systems with the ability to learn, adapt, and self-organize. These systems continuously evolve by redesigning their structure and behavior to best suit their context.

With recent advances in miniaturized microsystems, it is now possible to develop systems capable of sensing their environments and reacting accordingly. Research in smart materials is progressing toward developing systems with adaptive engineering functions and evolvable operational capabilities. These advances, including smart materials, nanotechnologies, communications, and AI, allow the development of smart systems that can react to their environments. For instance, advanced and smart materials/systems in a building can automatically respond and adapt to current and predicted situations such as occupancy, weather, and loads.

1.3 The Energy Efficient Building (EEB) Platform

1.3.1 Purpose and Strategic Importance

The Energy Efficient Building (EEB) Platform, synonymous with the EEBLAB, is a strategic initiative by UIR aimed at promoting sustainable building practices. The primary purpose of the EEB Platform is to develop and implement energy-efficient technologies and methodologies that can be applied to modern construction projects. This initiative is crucial for addressing the environmental challenges associated with urbanization and industrialization.



Figure 1.4: Energy Efficient Building (EEB) Platform

1.3.2 Contributions to Sustainable Solutions

The contributions of the EEB Platform to sustainable development are significant. By promoting energy-efficient building practices, the platform helps reduce the environmen-

tal impact of construction and supports the creation of greener, more sustainable urban environments. The research and innovations from the EEB Platform have the potential to revolutionize the construction industry, making it more sustainable and environmentally friendly. These contributions have been documented and disseminated through various publications, which can be accessed on the [EEB platform's publication page](#). More information about EEBLAB and its projects can be found on the official website: <https://eeb.uir.ac.ma/>.

1.3.3 Technologies and Methodologies

The EEB Platform leverages a range of innovative technologies and research methodologies. These include advanced insulation materials, energy-efficient HVAC systems, and smart building technologies that optimize energy use. The platform's research methodologies are designed to ensure that the solutions developed are both practical and scalable.

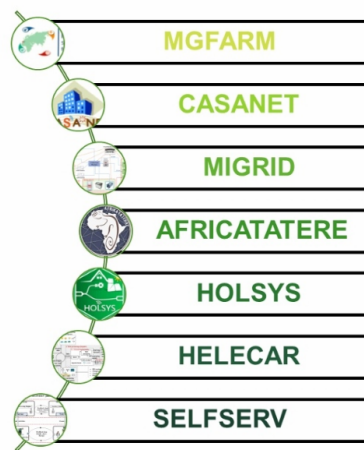


Figure 1.5: Key Projects under EEBLAB

1.4 Impact and Future Prospects

1.4.1 Influence on Research Landscape

UIR, through its various research initiatives and academic programs, has made a profound impact on the research landscape, both nationally and internationally. The university's contributions are recognized in numerous scholarly publications and have influenced the direction of research in several key areas.

1.4.2 Future Directions and Projects

Looking ahead, UIR is committed to continuing its tradition of excellence and innovation. The university has several ambitious projects in the pipeline, aimed at furthering its

research capabilities and expanding its academic offerings. These future directions reflect UIR's strategic goals of fostering a dynamic and impactful research environment that addresses the needs of society.

Conclusion

The International University of Rabat (UIR) has firmly established itself as a beacon of academic and research excellence in Morocco. Through its comprehensive academic programs and cutting-edge research initiatives, UIR continues to make significant contributions to science and technology, impacting both national and international research landscapes.

The LERMA Laboratory stands as a testament to UIR's dedication to innovation, driving forward the development of smart materials, systems, and technologies that address critical challenges in various sectors. The Energy Efficient Building (EEB) Platform further exemplifies UIR's commitment to sustainability and environmental stewardship. By promoting energy-efficient building practices and developing scalable, practical solutions, the EEB Platform plays a crucial role in fostering sustainable urban development.

As UIR looks to the future, it remains steadfast in its mission to blend tradition with modernity, shaping future leaders and pushing the boundaries of scientific and technological advancement. With ambitious projects on the horizon, UIR is poised to continue its trajectory of excellence, contributing to a smarter, more connected, and sustainable world.

Chapter 2

State of the Art

Introduction

Microgrids integrate renewable energy sources such as photovoltaic and wind systems with storage units, aiming to ensure reliable and efficient energy management. Their increasing penetration, driven by the rise of distributed energy resources (Distributed Energy Resources (DERs)), has highlighted the importance of advanced monitoring and control strategies. Recent literature has addressed embedded data acquisition (Data Acquisition (DAQ)) platforms for measuring key electrical parameters, supervisory control architectures for orchestration, and switching mechanisms for dynamic load management. Furthermore, the role of SCADA platforms and embedded systems has been emphasized for enabling real-time monitoring, data analytics, and decision support. Despite these advances, significant challenges persist. Among them are ensuring a balanced integration of heterogeneous sensors, enabling robust estimation of the State of Charge (SOC) of batteries, and supporting automated switching strategies that adapt to varying production and consumption profiles. These limitations underline the need for an intelligent, flexible system capable of both precise measurement and autonomous decision-making, which motivates the developments presented in this work.

2.1 Background and Context

2.1.1 Microgrids: Definition and Importance

A **microgrid** is a localized group of electricity sources and loads that can operate either connected to the main grid or independently in *islanded mode*. It integrates DERs such as solar panels, wind turbines, battery storage systems, and controllable loads within a defined boundary.

The importance of microgrids lies in their ability to:

- **Enhance reliability:** Microgrids can continue supplying power during grid outages by operating autonomously.
- **Increase efficiency:** By generating electricity close to the point of consumption, transmission losses are reduced.
- **Promote sustainability:** They enable the integration of renewable energy sources, reducing greenhouse gas emissions.
- **Support energy independence:** Communities, campuses, and industries can rely less on centralized grids.
- **Enable advanced energy management:** Through smart control and optimization, microgrids balance supply, demand, and storage in real time.

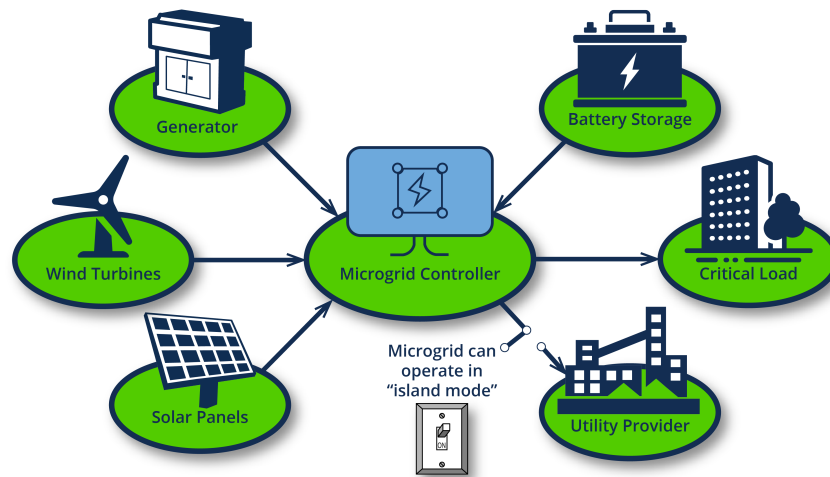


Figure 2.1: Schematic of a microgrid [1]

2.1.2 Renewable Energy Sources in Microgrids

Renewable Energy Sources (RES) play a central role in the development and operation of microgrids. Among the most widely deployed RES are Photovoltaic (PV) systems and Wind Turbines (WT). Their integration provides clean, sustainable, and often decentralized electricity production, which is essential for achieving the objectives of microgrids. However, their intermittent and weather-dependent nature introduces challenges for reliable power management.

2.1.2.1 Photovoltaic Systems

PV systems directly convert solar radiation into electrical energy through the *photovoltaic effect*. A PV cell generates a current when exposed to sunlight, and multiple cells connected in series and parallel form a solar panel. The performance of a PV system is

typically represented by its **current–voltage (I–V) characteristics**, which describe the nonlinear relationship between terminal voltage and output current under different irradiance and temperature conditions.

The current–voltage relationship of a PV cell can be described using the **single-diode model** as follows:

$$I = I_{ph} - I_0 \left(e^{\frac{q(V+IR_s)}{nkT}} - 1 \right) - \frac{V + IR_s}{R_{sh}} \quad (2.1)$$

where I is the output current, I_{ph} the photocurrent, I_0 the diode saturation current, q the electron charge, V the terminal voltage, R_s the series resistance, R_{sh} the shunt resistance, n the diode ideality factor, k the Boltzmann constant, and T the cell temperature in Kelvin.

The electrical power produced by the PV module is expressed as:

$$P = V \times I \quad (2.2)$$

and the **maximum power point (MPP)** is obtained when:

$$P_{max} = V_{mpp} \times I_{mpp} \quad (2.3)$$

where V_{mpp} and I_{mpp} are the voltage and current at the maximum power point.

The Maximum Power Point (MPP) of a PV module varies with solar irradiance and temperature, requiring control strategies such as *Maximum Power Point Tracking (MPPT)* to optimize energy extraction.

PWM vs. MPPT Controllers In stand-alone or grid-connected PV systems, charge controllers are used to regulate the power flow between the PV array, the batteries, and the loads. Two common methods are Pulse-Width Modulation (PWM) and Maximum Power Point Tracking (MPPT):

- **PWM:** A simpler and less expensive method, PWM controllers directly connect the PV array to the battery bank by rapidly switching the connection on and off. The charging voltage is essentially forced to match the battery voltage. While reliable, this method does not allow the PV system to operate at its maximum power point, which results in lower overall efficiency, especially when the PV voltage is much higher than the battery voltage.
- **MPPT:** MPPT controllers use DC-DC converters and advanced algorithms to continuously adjust the operating point of the PV array to track its maximum power point. This ensures maximum energy harvesting under varying irradiance and temperature conditions. Although MPPT controllers are more costly and complex, they

can increase energy yield by 20–30% compared to PWM controllers, particularly in systems where the PV array voltage significantly exceeds the battery voltage.

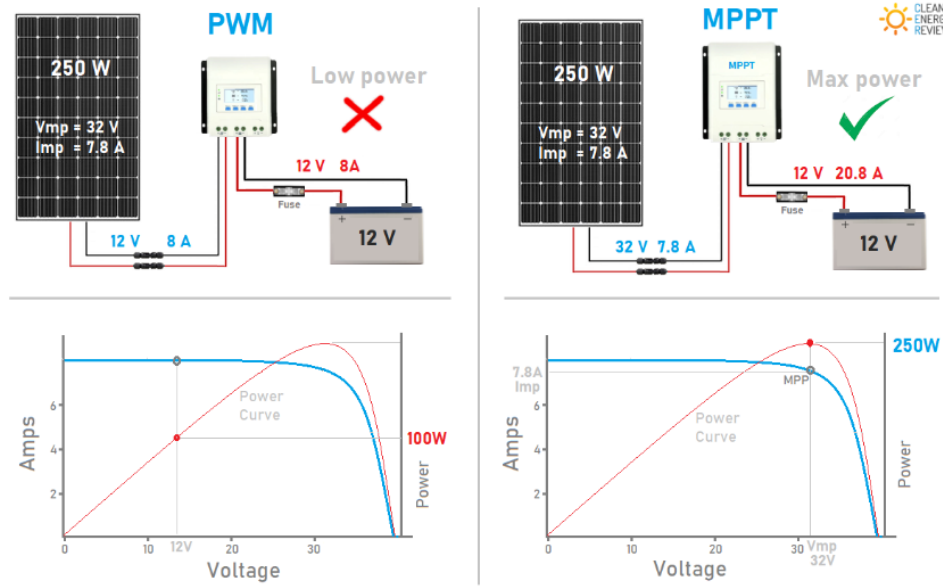


Figure 2.2: Comparison between PWM and MPPT charge controllers [2].

Despite their advantages, PV systems face inherent challenges:

- **Intermittency:** Output depends on solar irradiance, which varies with time of day, weather conditions, and seasonal changes.
- **Temperature dependence:** Higher cell temperatures reduce the conversion efficiency.
- **Integration needs:** To ensure stable operation in microgrids, PV systems often require energy storage and advanced power management.

2.1.2.2 Wind Turbines

Wind Energy Conversion System (WECS) transform the kinetic energy of wind into mechanical power via turbine blades, which is then converted into electrical power by a generator. The output power of a wind turbine is strongly dependent on the cube of wind speed, making accurate measurement and forecasting essential for integration into microgrids.

The theoretical power available in the wind is expressed as:

$$P_{wind} = \frac{1}{2} \rho A v^3 \quad (2.4)$$

where:

- ρ is the air density (kg/m^3),
- A is the swept area of the turbine blades (m^2),
- v is the wind speed (m/s).

The actual mechanical power extracted by a turbine is a fraction of this value, given by:

$$P_{\text{turbine}} = C_p \cdot \frac{1}{2} \rho A v^3 \quad (2.5)$$

where C_p is the power coefficient, limited by Betz's law to a theoretical maximum of 0.593 (59.3%).

There are two main types of wind turbines:

- **Horizontal-axis wind turbines (HAWT):** The most common design, suitable for large-scale applications with higher efficiency.
- **Vertical-axis wind turbines (VAWT):** Easier to install in decentralized environments but generally less efficient.



(a) Horizontal-Axis Wind Turbine (HAWT).



(b) Vertical-Axis Wind Turbine (VAWT).

Figure 2.3: Comparison between HAWT and VAWT designs.

While wind turbines contribute significantly to renewable energy generation, they also present limitations:

- **Variability:** Wind speed fluctuations lead to unpredictable power generation.
- **Cut-in and cut-out speeds:** Turbines only generate electricity within a certain wind speed range, limiting availability.

- **Environmental and spatial constraints:** Noise, visual impact, and site-specific wind profiles restrict deployment.

Overall, both PV systems and wind turbines are crucial elements in microgrids, but their intermittency necessitates the use of energy storage systems and intelligent control strategies to ensure reliability and balance between supply and demand.

2.1.3 Energy Storage Systems

2.1.3.1 Battery Technologies

Batteries are the most widely used energy storage devices in microgrids, enabling the balance between generation and demand, improving system reliability, and ensuring power quality. Among the different chemistries available, two dominant categories are commonly deployed:

- **Lead–acid batteries:** A mature and cost-effective technology, widely used in stand-alone and backup power systems. Their main advantages are low cost, robustness, and recyclability. However, they suffer from relatively low energy density, shorter cycle life, and reduced performance under deep discharge conditions.
- **Lithium–ion (Li-ion) batteries:** Characterized by high energy density, long cycle life, and high efficiency, Li-ion batteries have become the preferred choice for modern renewable energy systems and electric vehicles. Their disadvantages include higher cost and the need for sophisticated Battery Management Systems (BMS) to ensure safety and reliability.

Other chemistries such as nickel–metal hydride (NiMH), sodium–sulfur (NaS), or flow batteries are under development or applied in niche cases, but lead–acid and Li-ion remain the most relevant for microgrid applications.

2.1.3.2 Battery State of Charge SOC

The **State of Charge SOC** is defined as the ratio between the current capacity of the battery and its nominal (maximum) capacity, usually expressed as a percentage:

$$SOC(t) = \frac{Q(t)}{Q_{nom}} \times 100\% \quad (2.6)$$

where $Q(t)$ is the remaining charge in the battery at time t , and Q_{nom} is the nominal capacity under specified conditions.

SOC plays a crucial role in microgrid management, since it provides an indication of how much energy is available for supplying loads or absorbing excess renewable generation. Accurate SOC information is essential for:

- **Energy management:** Ensuring optimal charging and discharging strategies to maximize battery lifetime and system efficiency.
- **System reliability:** Preventing overcharge and deep discharge that could lead to safety issues or premature degradation.
- **Integration with renewables:** Allowing intelligent coordination with intermittent sources such as PV and wind to maintain supply–demand balance.

2.1.4 SOC Estimation Methods

Accurate estimation of the battery SOC is essential for effective energy management in microgrids and renewable energy systems. Since SOC cannot be measured directly, it must be estimated from measurable variables such as voltage, current, and temperature, often combined with mathematical models. Several approaches have been developed, each with different levels of accuracy, complexity, and implementation cost.

2.1.4.1 Voltage-Based Estimation

The voltage-based method relies on the direct relationship between the battery terminal voltage and its SOC. In open-circuit conditions, the **Open-Circuit Voltage (OCV)** of the battery can be mapped to SOC through experimental calibration curves.

$$SOC \approx f(V_{OCV}) \quad (2.7)$$

where $f(\cdot)$ is a nonlinear mapping obtained from battery characterization tests.

Advantages:

- Simple and low-cost implementation.
- Requires only voltage measurement.

Limitations:

- Accuracy is poor under load because terminal voltage is affected by internal resistance and transient dynamics.
- Requires long rest periods to reach open-circuit conditions.

2.1.4.2 Coulomb Counting

The Coulomb counting method estimates SOC by integrating the battery current over time. It is based on the charge balance principle:

$$SOC(t) = SOC(t_0) - \frac{1}{Q_{nom}} \int_{t_0}^t I(\tau) d\tau \quad (2.8)$$

where $I(\tau)$ is the instantaneous current (positive during discharge, negative during charge) and Q_{nom} is the nominal capacity.

Advantages:

- Works in real time under dynamic load conditions.
- Does not require battery rest periods.

Limitations:

- Accumulates integration errors due to current sensor drift.
- Requires accurate knowledge of initial SOC.
- Sensitive to variations in Q_{nom} due to aging and temperature.

2.1.4.3 Model-Based and AI Approaches

To overcome the limitations of simple methods, advanced estimation techniques combine battery models with state estimation algorithms or data-driven approaches.

Kalman Filter-Based Methods The battery is represented by an Equivalent Circuit Model (ECM), such as the *Thevenin model*, which captures voltage dynamics through resistors and capacitors. The state-space form can be expressed as:

$$x_{k+1} = Ax_k + Bu_k + w_k \quad (2.9)$$

$$y_k = Cx_k + Du_k + v_k \quad (2.10)$$

where x_k represents the state vector (including SOC), u_k is the input (current), y_k is the measured output (voltage), and w_k, v_k are process and measurement noise, respectively.

The Extended Kalman Filter (EKF) or Unscented Kalman Filter (UKF) is then applied to recursively estimate SOC.

Neural Networks and Machine Learning Artificial intelligence (AI) approaches rely on experimental data such as voltage, current, temperature, and SOC to build models able to predict SOC even under complex and uncertain conditions. Instead of using only equations, these methods learn directly from data. Different types of models can be used, ranging from simple ones that only consider the present state, to more advanced ones that can also take into account the history of battery use. In some cases, Artificial Intelligence (AI) is also combined with traditional physical models to improve accuracy.

Advantages of AI-based methods:

- They can reach high accuracy even under variable operating conditions.

- They are able to consider the effects of temperature changes and battery aging.
- They are generally more robust when measurements are noisy or uncertain.

Drawbacks:

- They require more computational resources compared to simple methods.
- They need a large amount of good-quality data for training.
- Setting up and adjusting these models can be complex and depend on the specific battery system.

2.1.4.4 Comparison and Limitations

Despite the progress of advanced techniques, SOC estimation remains challenging. Key limitations include:

- **Battery variability:** Capacity and voltage characteristics change over lifetime due to aging.
- **Environmental effects:** Temperature strongly affects voltage, internal resistance, and efficiency.
- **Nonlinearity:** The SOC–voltage relationship is nonlinear and different across chemistries (Lead–acid vs Li-ion).
- **Trade-off:** Simpler methods are fast and inexpensive but inaccurate, while advanced methods require computation and calibration.

In practice, hybrid approaches are often used, combining Coulomb counting for dynamic tracking with OCV-based calibration or Kalman filters to correct integration drift. This provides a balance between accuracy and real-time applicability.

2.1.5 Embedded Systems for Energy Monitoring

Embedded systems play a central role in monitoring and controlling microgrids. Low-cost platforms such as Arduino and Raspberry Pi are widely used because they can easily interface with a variety of sensors.

- **Arduino boards** are well-suited for real-time data acquisition, such as measuring voltage, current, temperature, and irradiance from sensors.
- **Raspberry Pi boards** provide higher computing power and network capabilities, making them suitable for data storage, local processing, and integration with web servers or cloud platforms.

- **Sensors** are essential to capture key parameters of the microgrid: current sensors (e.g., Hall effect), voltage dividers, temperature sensors, pyranometers for solar radiation, and anemometers for wind speed.

These embedded systems act as the interface between the physical microgrid and the digital monitoring and control layer.

2.1.6 Supervisory Control and Data Acquisition (SCADA) for Microgrids

Supervisory Control and Data Acquisition (SCADA) systems are used to monitor, visualize, and control distributed energy resources in a microgrid. They provide dashboards, data logging, and decision-making support.

- **Node-RED** is a flow-based tool for connecting devices, sensors, and APIs. It allows fast prototyping of monitoring systems.
- **ThingsBoard** is an open-source IoT platform that offers dashboards, alarms, and device management, making it well-suited for large-scale microgrid supervision.
- **Communication protocols:**
 - **MQTT** (Message Queuing Telemetry Transport) is lightweight and efficient, ideal for real-time sensor data transmission.
 - **HTTP** is simpler and widely used, but less efficient for continuous, real-time monitoring.

In practice, MQTT is often preferred for microgrids due to its low bandwidth consumption and reliability.

2.1.7 Switching and Load Management in Microgrids

Efficient switching and load management are crucial to balance generation, storage, and consumption in microgrids.

- **Relays and contactors** are used to switch loads or connect/disconnect different energy sources.
- **Automatic transfer systems** ensure seamless switching between the main grid and backup sources (e.g., batteries or generators).
- **SOC-based switching:** The State of Charge (SOC) of the battery is often used as a decision criterion. For example:

- If SOC is high, non-critical loads can be connected.
- If SOC is low, only priority loads are maintained, and renewable sources or the main grid take over.

This intelligent switching increases the lifetime of the battery, avoids over-discharging, and ensures reliable supply for critical loads.

2.2 Literature Review

2.2.1 Renewable Energy Sources in Microgrids

Hybrid microgrids often rely on the combined use of PV systems and wind turbines, as their complementary generation profiles can reduce the overall intermittency of renewable production. Several studies highlight that PV is typically the dominant contributor in daytime hours, whereas wind can extend availability during night or cloudy conditions [3], [4].

Nevertheless, both PV and wind introduce significant variability. Solar irradiance depends strongly on diurnal and seasonal cycles, as well as transient cloud cover, while wind generation is governed by stochastic fluctuations in wind speed. This intermittency has been shown to complicate forecasting, system sizing, and real-time balancing [5], [6]. Consequently, hybridization of multiple sources together with storage is a widely investigated strategy to mitigate power fluctuations.

Monitoring of these sources is essential to enable adaptive control and accurate performance assessment. Typical approaches include:

- **Solar irradiance sensors** (pyranometers or reference cells) to correlate solar availability with PV production.
- **Anemometers** for wind speed measurement, often complemented by wind vane orientation to characterize turbine conditions.
- **Electrical measurements** of current and voltage, used to reconstruct I–V curves for PV modules or to calculate instantaneous power at the wind turbine generator terminals.

Such monitoring infrastructures are fundamental inputs for MPPT, predictive energy management, and diagnostic analysis.

2.2.2 Energy Storage and SOC-Based Management

Batteries remain the backbone of microgrid operation, ensuring continuity of supply and smoothing renewable intermittency. Research emphasizes their dual role as both a buffer

and a stabilizer, enabling local autonomy and improving reliability during grid disturbances [7], [8].

The SOC has emerged as a pivotal variable for energy management. It reflects the available capacity in real time and is thus central for balancing production and consumption. A large body of work shows that SOC-driven strategies enhance the utilization of renewables, extend battery lifetime, and support priority-based load allocation [9], [10].

In particular, literature has explored SOC-based switching mechanisms to determine the operating mode of microgrids. For example, thresholds can be set to activate load shedding, trigger inter-microgrid energy transfers, or switch between grid-connected and islanded modes [11], [12]. More advanced approaches integrate predictive control or optimization techniques, where SOC serves as a state variable in decision-making models.

Energy sharing between interconnected microgrids is another active research direction. Here, SOC information enables cooperative strategies: surplus energy from one microgrid can be exported when its batteries are near full capacity, while neighboring systems with depleted SOC values can import power to maintain stability [13], [14]. These works underline SOC as a critical enabler for distributed cooperation and resilient microgrid networks.

2.2.3 Embedded Systems for Data Acquisition

The acquisition of electrical parameters in microgrids requires dedicated embedded platforms capable of measuring voltage, current, and power with sufficient accuracy. Microcontrollers such as **Arduino**, **ESP32**, and **STM32** have been widely adopted in academic and industrial prototypes due to their low cost, flexibility, and large ecosystem of sensors [15], [16].

Several studies report the design of **PCB-based systems** for data acquisition, integrating current sensors (e.g., Hall-effect) and voltage dividers to monitor photovoltaic modules, wind turbines, and battery parameters [17], [18]. Compared to commercial data loggers, these custom boards offer lower cost and can be tailored to specific applications in hybrid microgrids.

Wireless communication modules, such as **Bluetooth**, **ZigBee**, or Wi-Fi, are frequently integrated to transmit measurements to a supervisory device [19], [20]. In particular, Bluetooth has been applied for point-to-point data exchange with local controllers, while Wi-Fi and ZigBee enable multi-node acquisition networks. Such architectures reduce wiring complexity and allow **distributed sensing** across geographically separated renewable energy sources.

2.2.4 Data Processing and Local Control Units

Once acquired, raw data must be processed locally to extract useful indicators such as power, energy balance, and SOC. For this purpose, embedded computers like the **Raspberry Pi** have emerged as powerful *edge-computing devices* [21]. Compared to micro-controllers, they provide higher computational capacity, multi-tasking capabilities, and direct connectivity to IoT platforms.

Recent research highlights the use of **Python scripts** on Raspberry Pi for the real-time calculation of electrical parameters, SOC estimation, and local decision-making [22]. This software-based approach offers flexibility in implementing algorithms ranging from simple power balance to advanced control strategies.

Distributed control architectures have been widely explored in the literature [23], [24]. In these systems, local controllers process measurements independently and cooperate with supervisory layers for coordination. This paradigm enhances reliability, as the failure of a central unit does not compromise the entire microgrid. Moreover, distributed controllers enable faster local reactions (e.g., switching decisions based on SOC) while still exchanging information with higher-level platforms.

2.2.5 Supervisory Control and Communication in Microgrids

Supervisory systems in microgrids collect, visualize, and orchestrate data coming from distributed DERs and storage units. They are generally based on Supervisory Control and Data Acquisition (SCADA) architectures, which provide acquisition, historical logging, alarms, and operator interfaces [25], [26].

For local orchestration and rapid prototyping, tools such as **Node-RED** are widely employed. They allow the assembly of data flows, integration of sensors/actuators, and routing of messages toward higher layers (databases, cloud platforms). Several studies demonstrated the effectiveness of Node-RED for IoT applications and decentralized orchestration architectures [27], [28].

At the protocol level, the comparison between Message Queuing Telemetry Transport (MQTT) and Hypertext Transfer Protocol (HTTP) is particularly relevant for connected energy systems. MQTT is a lightweight publish/subscribe protocol, designed for low-bandwidth communications and intermittent connectivity, making it a common choice for real-time telemetry [29]. HTTP, by contrast, is simple and universal (REST), but less efficient for continuous real-time streams and for handling a large number of frequent, small messages [30].

For supervision and visualization, platforms such as **ThingsBoard** offer ready-to-use interfaces and support multiple transports (MQTT, HTTP, CoAP). They enable device management, dashboards, rule engines, and remote commands [31]. Integration between Node-RED and ThingsBoard is a common architecture to combine local orchestration

with cloud-based supervision.

2.2.6 Switching and Load Management Strategies

Switching and load management strategies are central to ensuring service continuity and protecting storage units. Hardware solutions range from simple relays and contactors to more advanced Automatic Transfer Switch (ATS) capable of transferring loads between sources (grid, PV, diesel generator, batteries) according to predefined rules [32], [33].

The literature generally distinguishes:

- **Manual strategies**, where an operator triggers the switching;
- **Automatic strategies**, based on thresholds (voltage, frequency, SOC, load priority);
- **Assisted/fast strategies**, which combine local sensors and supervisory layers to enable seamless transfer without noticeable interruption.

Recent studies propose SOC-driven approaches for battery balancing and load transfer decisions between microgrids. These methods evaluate the SOC of each bank and determine transfer directions (or shedding actions) to avoid deep discharge and extend battery lifetime [34], [35].

Multi-mode systems (Normal, Assistance, Emergency) are also widely reported. In Normal mode, power flows follow nominal configuration. In Assistance mode, a backup source supports priority loads. In Emergency mode, the system switches to survival strategies (shedding, isolation) to preserve the minimum SOC. Such modes are implemented through embedded logic and actuators (relays, contactors) driven by local controllers or orchestrators such as Node-RED/SCADA [26], [32].

2.2.7 Summary and Research Gap

The literature has demonstrated significant progress in the design of embedded systems, communication protocols, and control architectures for Microgrid (MG) applications. Researchers have successfully developed low-cost DAQ boards [17], [18], wireless acquisition systems [19], [20], and distributed controllers [23], [24]. These works highlight the feasibility of combining microcontrollers, embedded computers, and Internet of Things (IoT) platforms for real-time monitoring and control. Furthermore, comparative studies of communication protocols such as MQTT and HTTP [25], [30] have emphasized their suitability for scalable DERs integration.

Despite these advancements, several limitations persist. First, most acquisition systems focus on either photovoltaic or wind subsystems, while the integration of multiple

renewable sources within a unified DAQ platform remains limited [26]. Second, existing switching strategies are often based on fixed thresholds, with limited consideration of real-time SOC-driven decision making. Finally, while IoT platforms like ThingsBoard and Node-RED facilitate data orchestration, their integration with fully automated load management strategies is still underexplored.

The present work aims to address these gaps by proposing a microgrid-oriented embedded platform capable of multi-sensor integration (PV, wind, and batteries), real-time SOC-based switching, and seamless connectivity with IoT platforms for monitoring and control. This contribution bridges the gap between academic prototypes and practical deployment scenarios, enhancing both flexibility and reliability of MG operations.

2.3 Problem Statement and Objectives

2.3.1 Problem Statement

The growing deployment of renewable energy sources such as photovoltaic (PV) panels and wind turbines has opened new opportunities for powering isolated zones and rural communities. However, the integration of these intermittent sources within microgrids remains a major challenge. The main limitations encountered in existing solutions can be summarized as follows:

- **Unreliable energy availability:** Renewable energy production is variable and often insufficient to ensure continuous supply without efficient energy management strategies.
- **Lack of intelligent supervision and automation:** Many small-scale systems rely on manual monitoring and fixed switching logic, which cannot adapt dynamically to variations in production and demand.
- **Absence of energy sharing between microgrids:** Existing commercial systems are generally designed for isolated single-microgrid operation. The concept of energy exchange between two or more microgrids in rural areas is rarely addressed, despite its potential to increase reliability and self-sufficiency.
- **Non-open-source and costly solutions:** Commercial platforms (e.g., Victron, Schneider) offer advanced management functions, but they remain expensive, proprietary, and difficult to customize for research or low-cost rural applications.
- **Limited use of SOC-based decision-making:** Battery management is often based on simple voltage thresholds, which do not provide accurate state estimation. This results in inefficient switching decisions and possible degradation of storage systems.

These limitations highlight the need for a system that combines **real-time monitoring, SOC-based control, and open-source supervision**, while allowing **energy sharing between interconnected microgrids**.

Conventional Isolated Microgrid

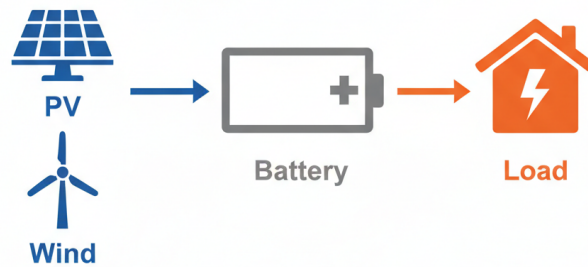


Figure 2.4: Conventional isolated microgrid without SOC-based management.

2.3.2 Motivation

The motivation behind this work is to design and implement a system that addresses the limitations above by integrating advanced control strategies and open-source technologies. Specifically:

- To test advanced SOC-based switching strategies on a real microgrid demonstrator.
- To develop an open-source, low-cost platform that combines monitoring, supervision, and automation in one coherent framework.
- To explore the concept of interconnected microgrids, where two or more microgrids can share energy based on the state of charge (SOC) of their respective batteries.
- To provide a laboratory-scale platform for research and education, enabling testing of different algorithms, communication protocols, and energy management strategies.
- To demonstrate that such a system, once scaled up, could be adapted to rural electrification and isolated zones, where reliable and affordable energy management solutions are urgently needed.

2.3.3 Project Objectives

The specific objectives of the project can be outlined as follows:

1. **Real-time monitoring of renewable energy sources and storage systems**
 - Measure PV and wind turbine voltage, current, and power.
 - Monitor battery voltages and bidirectional currents (charging and discharging).
2. **Accurate SOC estimation and energy balance calculations**
 - Implement algorithms for SOC calculation of each battery.
 - Compute instantaneous and cumulative energy flows within and between microgrids.
3. **Automatic SOC-based switching between microgrids**
 - Develop a control logic that dynamically switches loads and energy sharing between microgrids according to SOC levels.
 - Ensure continuous energy availability while protecting battery health.
4. **Integration of monitoring, supervision, and automation**
 - Centralize data acquisition and control decisions using a Raspberry Pi.
 - Use Node-RED for communication between hardware boards and higher-level applications.
 - Implement remote supervision and control via the ThingsBoard IoT platform.
5. **Data analysis and validation**
 - Correlate solar power production with solar irradiation measurements.
 - Correlate wind turbine production with wind speed and orientation.
 - Validate switching logic and system performance through laboratory testing.
6. **Proof of concept for research and rural applications**
 - Provide a laboratory-scale demonstrator suitable for research activities.
 - Establish a foundation for future adaptation to isolated zones and rural electrification projects.

Conclusion

In summary, the study of microgrids reveals a strong foundation of renewable integration, storage solutions, and intelligent control strategies. Considerable progress has been achieved in DAQ platforms, supervisory systems, and distributed control methods. However, current solutions still face limitations in real-time SOC-based switching, seamless sensor integration, and hybrid communication frameworks that ensure scalability and robustness. These gaps highlight the need for a comprehensive solution that unifies monitoring, communication, and intelligent decision-making. Building upon this state of the art, the next chapter will focus on the design of the proposed system, addressing these challenges through an integrated architecture tailored for reliable microgrid operation.

Chapter 3

System Design

Introduction

This chapter presents the design of the proposed dual-microgrid system. It details the overall architecture, hardware components, software layers, communication framework, and control logic, providing a foundation for implementation and testing.

3.1 Overall System Architecture

3.1.1 Concept of Dual Microgrids

3.1.1.1 Description of Microgrid 1 (MG1)

Microgrid 1 (**MG1**) is a small-scale renewable energy system integrating both wind and solar sources to supply a local load. It consists of a **wind turbine**, a **solar panel**, a **hybrid charge controller**, a **rechargeable battery**, and **Load 1**, a fan-based ventilation system controlled according to CO₂ levels. The system provides a laboratory-scale platform for testing SOC-based energy management and energy sharing between distributed sources.

Wind Turbine MG1 is equipped with a **Silent Wind 12V wind turbine** (Permanent Magnet Generator, 3-phase AC) rated at 12 VDC, 420 W. The turbine starts producing energy at 2.2 m/s and reaches peak power at 14.5 m/s.

Table 3.1: Specifications of Silent Wind 12V Turbine [36]

Parameter	Value
Rated Voltage	12 VDC
Rated Power	420 W
Start-up Speed	2 m/s
Charging Start Speed	2.2 m/s
Peak Power Wind Speed	14.5 m/s
Rotor Diameter	1.15 m
Number of Blades	3
Blade Weight	150 g per blade
Blade Material	Carbon fiber, hand laminated
RPM Range	550 – 1700 rpm
Generator Weight	6.8 kg
Package Dimensions	780 × 400 × 210 mm
Package Weight	10 kg
Color	White RAL 9010, powder coated



Figure 3.1: Silent Wind 12V wind turbine [37]

Solar Panel The solar component is a **SL50CE-18 solar panel** by NOWER (Japan Technology) providing 50 W maximum power. Key parameters are shown in Table 3.2.

Table 3.2: Specifications of SL50CE-18 Solar Panel [38]

Parameter	Value
Maximum Power (P_m)	50 W
Open-Circuit Voltage (V_{oc})	21.92 V
Short-Circuit Current (I_{sc})	2.96 A
Maximum Power Voltage (V_{mp})	18 V
Maximum Power Current (I_{mp})	2.77 A
Operating Temperature	45°C to +85°C
Tolerance	$\pm 3\%$



Figure 3.2: SL50CE-18 Solar Panel [38]

Hybrid Charge Controller A Silent Wind 12V hybrid charge controller manages both PV and wind inputs. It supports 40 A maximum for wind, 20 A maximum for PV, and provides 2×10 A DC output. It protects batteries from overcharge and reverse current flow.



Figure 3.3: Silent Wind 12V Hybrid Charge Controller [39]

Battery Energy storage is provided by a **12V/150Ah sealed lead-acid (gel) battery** from OUTDO. It ensures continuous supply to Load 1 and is compatible with the hybrid charge controller.



Figure 3.4: OUTDO 12V/150Ah Gel Battery [40]

Load 1 Load 1 is a **ventilation system** powered by the battery, operating at 12 VDC. It maintains indoor air quality based on CO₂ levels, demonstrating practical usage and testing of SOC-based energy management.



Figure 3.5: Load 1: CO₂-controlled ventilation system [41]

3.1.1.2 Description of Microgrid 2 (MG2)

Microgrid 2 (MG2) is designed as a photovoltaic-based energy system with storage and a DC load. It integrates a high-power PV panel, an advanced MPPT charge controller, a gel battery for storage, and a low-power refrigeration system as Load 2. This configuration allows testing of PV-only microgrids and their interaction with other renewable-based systems.

Photovoltaic Panel MG2 is equipped with an **AS-6P30-265W solar panel** manufactured by Amerisolar. This module delivers a maximum power output of 265 W, making it suitable for medium-scale off-grid applications.

Electrical Specifications:

- Maximum Power (P_m): 265 W
- Open-Circuit Voltage (V_{oc}): 37.6 V
- Short-Circuit Current (I_{sc}): 9.13 A
- Maximum Power Voltage (V_{mp}): 30.4 V
- Maximum Power Current (I_{mp}): 8.72 A
- Module Efficiency: 16.2%
- Operating Temperature Range: -40°C to $+85^{\circ}\text{C}$ [42]



Figure 3.6: Amerisolar AS-6P30-265W photovoltaic panel [42]

Charge Controller The PV output is managed by a **Victron BlueSolar MPPT 150/35 charge controller**. This advanced controller uses maximum power point tracking (MPPT) to optimize energy extraction from the PV panel under varying irradiation conditions.

Specifications:

- Nominal battery voltage: 12 V, 24 V, 36 V, 48 V (auto-select)
- Maximum PV open circuit voltage: 150 V

- Rated charge current: 35 A
- Battery charging algorithm: Multi-stage (bulk, absorption, float, equalization)
- Communication: VE.Direct port for monitoring and integration[43]



Figure 3.7: Victron BlueSolar MPPT 150/35 Charge Controller [43])

Battery Energy storage for MG2 is provided by a **Victron Energy 12V 130Ah GEL battery**. This deep-cycle gel battery is maintenance-free and designed for long service life, making it ideal for renewable energy storage applications.

Specifications:

- Nominal voltage: 12 V
- Capacity (C10): 130 Ah
- Technology: GEL, deep cycle
- Maintenance: Sealed, maintenance-free
- Application: Off-grid solar, backup, renewable integration



Figure 3.8: Victron Energy 12V/130Ah GEL Battery [44]

Load 2 The load of MG2 is an **Atlantic portable refrigerator** designed for car use, operating at 12 V. With a capacity of 24 liters, it ensures cooling and storage of food or medicine in off-grid scenarios. The consumption is between 3 A and 4 A, corresponding to an estimated power of approximately 42 W.



Figure 3.9: Load 2: Atlantic 12V Portable Refrigerator

3.1.2 Block Diagram of the Proposed System

Figure 3.10 illustrates the general architecture of the two microgrids (MG1 and MG2) and their interconnection link. The diagram highlights the energy flow, control signals, and communication between components.

The system is structured into three main aspects:

- **Measurement:** acquisition of electrical parameters (voltage, current, SOC) through PCBs and sensors.
- **Control:** switching and energy management decisions executed by microcontrollers (Arduino and Raspberry Pi).
- **Supervision:** monitoring, visualization, and data logging using Node-RED and ThingsBoard.

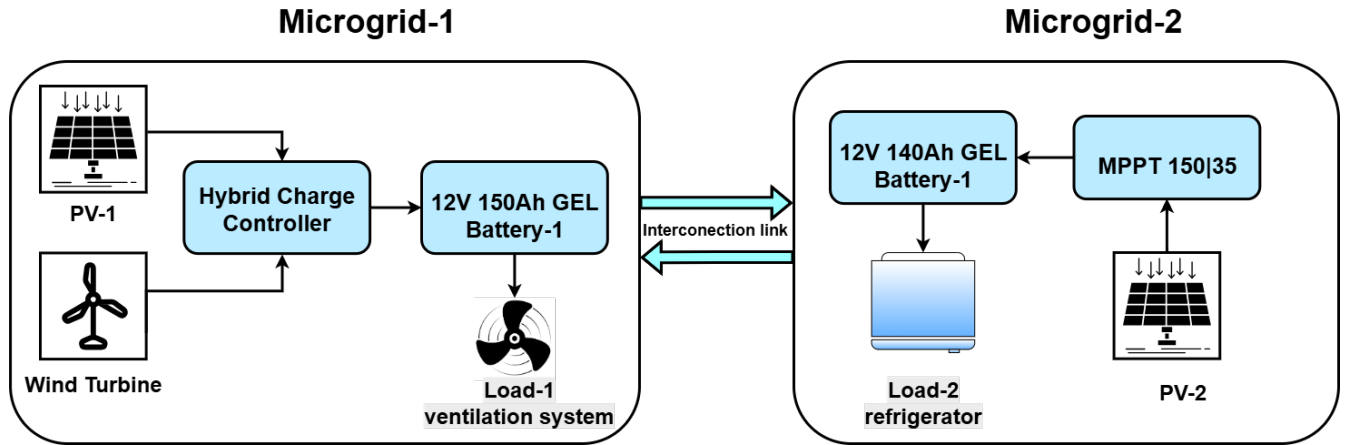


Figure 3.10: General architecture of MG1 and MG2 with interconnection link.

3.1.3 Functional Layers of the System

The proposed system is organized into four functional layers, each representing a specific role in the microgrid operation:

Energy Layer This layer consists of the primary energy sources and storage units, including the photovoltaic panels, wind turbine, rechargeable batteries, and connected loads. It ensures the physical energy conversion and supply to the end applications.

Measurement Layer The measurement layer integrates the dedicated PCBs with voltage dividers, ACS current sensors, and data acquisition circuits. These components provide real-time monitoring of system states such as voltage, current, and state-of-charge (SOC).

Control Layer The control layer is implemented using Arduino microcontrollers and Raspberry Pi boards. It executes switching logic, SOC-based energy management algorithms, and ensures proper operation of the dual-microgrid system.

Supervision Layer At the top level, the supervision layer enables remote monitoring, visualization, and interaction with the system. Node-RED manages local communication and flow control, while ThingsBoard provides a cloud-based dashboard for data supervision and reporting.

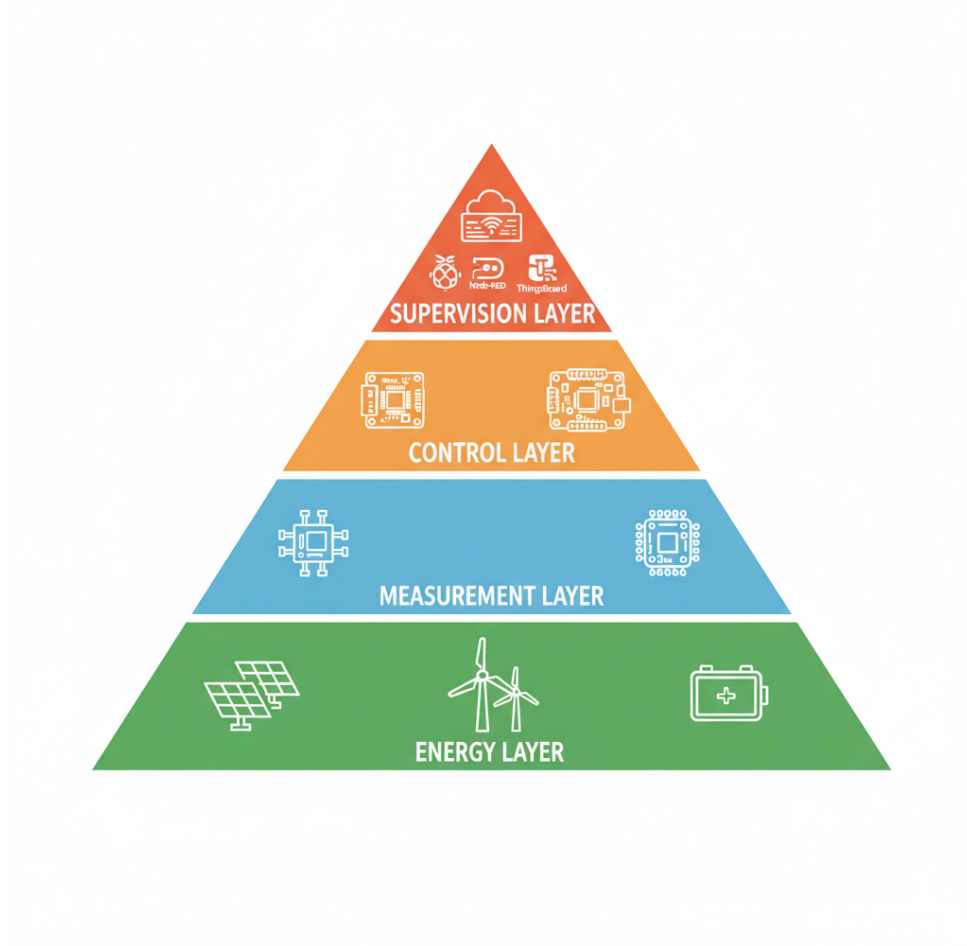


Figure 3.11: Triangular representation of the functional layers

3.2 Hardware Design

3.2.1 Measurement Circuits

The measurement circuits are developed on custom-designed PCBs to acquire real-time voltage and current data from different parts of the dual microgrid system. These circuits are based on the **Arduino Nano** microcontroller for data acquisition and processing, and the **ACS712 current sensor** for current measurement.

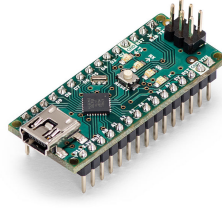


Figure 3.12: Arduino Nano used as the main acquisition unit [45]

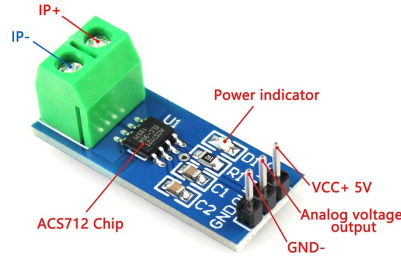


Figure 3.13: ACS712 current sensor module

3.2.1.1 PCB1: PV2 Measurement

Components: Arduino Nano, ACS712 current sensor, voltage divider circuit (with $R_1 = 74 \text{ k}\Omega$ and $R_2 = 10 \text{ k}\Omega$), and a Zener diode for protection.

Function: This PCB is designed to measure the **voltage and current of PV2**.

- The **ACS712 sensor** provides the current measurement. It outputs an analog voltage proportional to the bidirectional current flowing through the panel.
- The **voltage divider** scales down the PV voltage from its maximum open-circuit value ($V_{oc} \approx 22 \text{ V}$ for PV2) to a safe level for the Arduino Nano's ADC pin, which is limited to $0 - 5 \text{ V}$.
- A **Zener diode** is added across the Arduino input pin to clamp any overvoltage, ensuring protection against unexpected transients or spikes.

Voltage Divider Design: The voltage divider is composed of resistors $R_1 = 74 \text{ k}\Omega$ (connected to the PV positive terminal) and $R_2 = 10 \text{ k}\Omega$ (connected to ground). The output voltage applied to the Arduino input is given by:

$$V_{\text{out}} = V_{\text{in}} \times \frac{R_2}{R_1 + R_2}$$

Substituting the resistor values:

$$V_{\text{out}} = V_{\text{in}} \times \frac{10}{74 + 10} = V_{\text{in}} \times \frac{10}{84}$$

$$V_{\text{out}} \approx 0.119 \times V_{\text{in}}$$

Thus, for a maximum PV input voltage of 38 V:

$$V_{\text{out}} \approx 38 \times 0.119 \approx 4.522 \text{ V}$$

This ensures the Arduino safely measures the PV voltage within its ADC input range (0–5 V).

Zener Diode Protection: A 5.1 V Zener diode is connected in parallel with the Arduino input pin. In case of abnormal conditions (e.g., PV surge voltage, faulty resistor divider, or noise spikes), the Zener diode clamps the voltage to a safe maximum of $\approx 5.1 \text{ V}$. This prevents potential damage to the ADC input and guarantees circuit robustness.

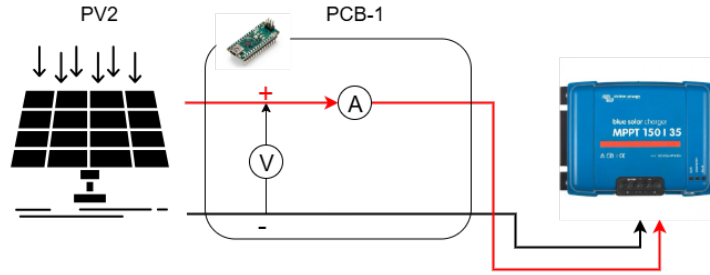


Figure 3.14: PCB1 Architecture for PV2 measurement.

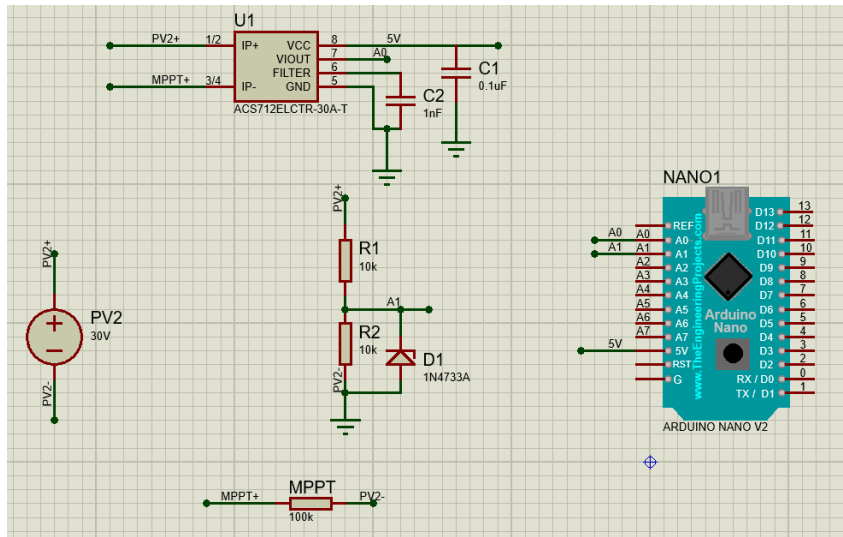


Figure 3.15: PCB1 schematic for PV2 measurement

3.2.1.2 PCB2: Battery Monitoring

Components: Arduino Nano, four ACS712 current sensors, and two voltage divider circuits.

Function: PCB2 is designed to monitor the **charging and discharging behavior of Battery 1 and Battery 2**. For each battery, two ACS712 sensors are used:

- One sensor measures the input (charging) current.
- The other sensor measures the output (discharging) current.

In total, four ACS712 sensors are implemented for both batteries. Additionally, two voltage divider modules (rated for $< 25\text{ V}$) are used to measure the voltages of the two batteries, scaling them down to safe levels for the Arduino ADC.

Note on Design Choice: Although the ACS712 is capable of bidirectional current detection, in this design we opted to use two separate sensors per battery. This approach simplifies data processing in the Arduino, since one sensor is dedicated to charging and the other to discharging, eliminating the need to interpret current direction.

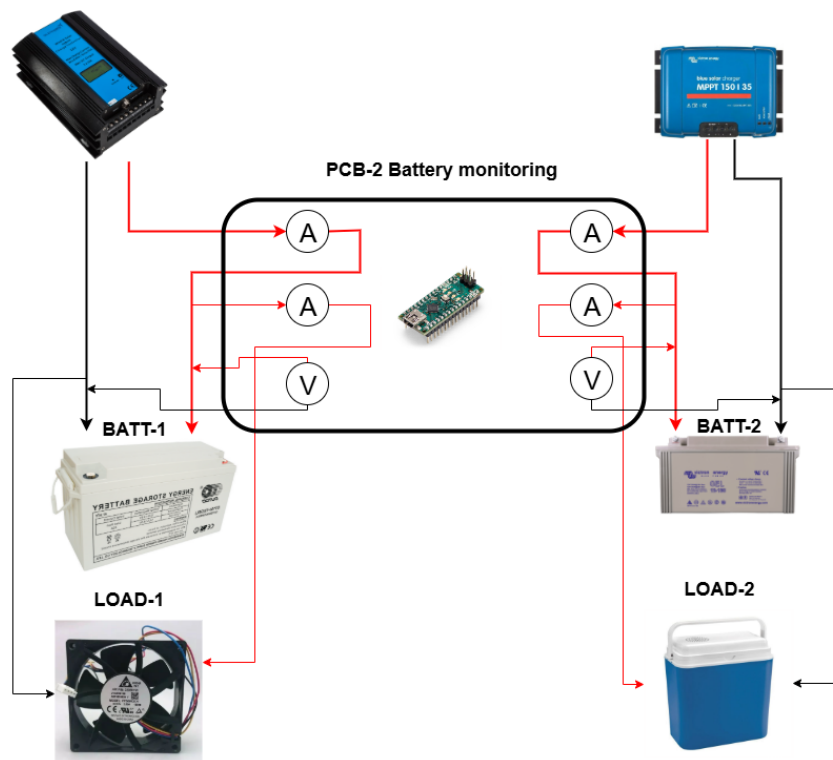


Figure 3.16: PCB2 Arcitecture for monitoring Battery 1 and Battery 2.

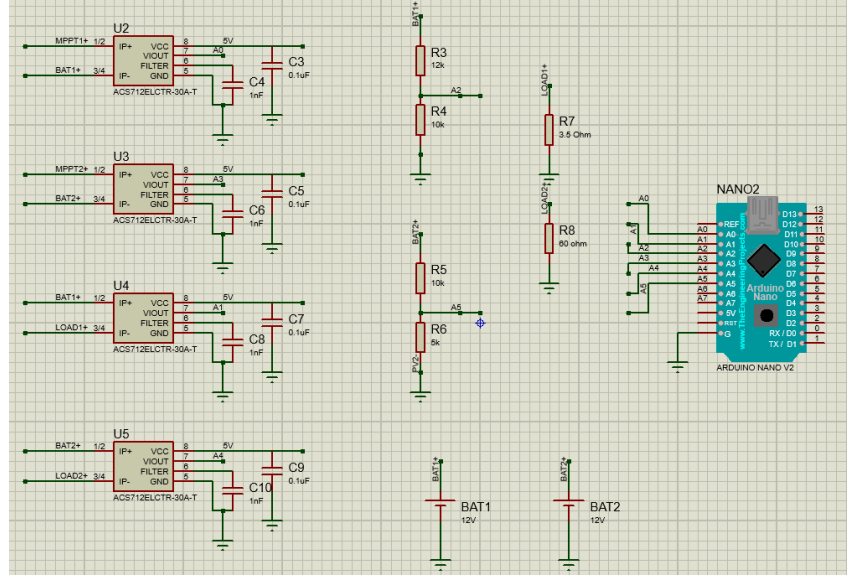


Figure 3.17: PCB2 schematic for monitoring Battery 1 and Battery 2.

3.2.1.3 Data Acquisition from MG1 Sources

Communication Setup: Data from MG1 (PV1 and wind turbine measurements) is transmitted via **Bluetooth communication** between a secondary Raspberry Pi connected to the hybrid charge controller and the main Raspberry Pi of MG2.

Data Flow:

- Hybrid charge controller provides voltage and current measurements of PV1 and wind turbine.
- Secondary Raspberry Pi collects this data via Bluetooth.
- The data is transmitted using **MQTT protocol** to the main Raspberry Pi through **Node-RED**, where it is integrated with MG2 measurements.

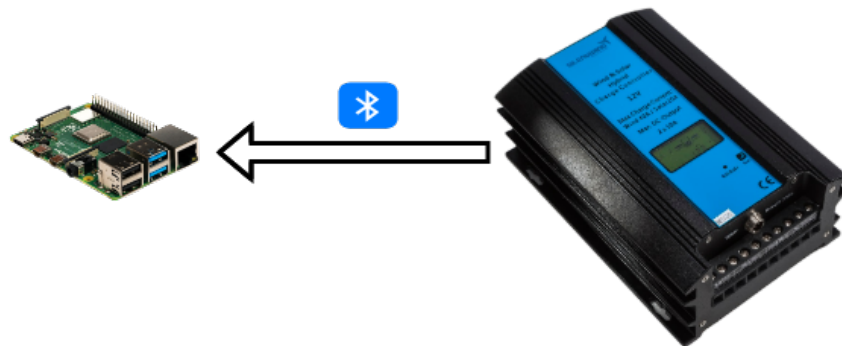


Figure 3.18: Bluetooth-based data acquisition from MG1 sources .

3.2.2 Switching Circuit (PCB3)

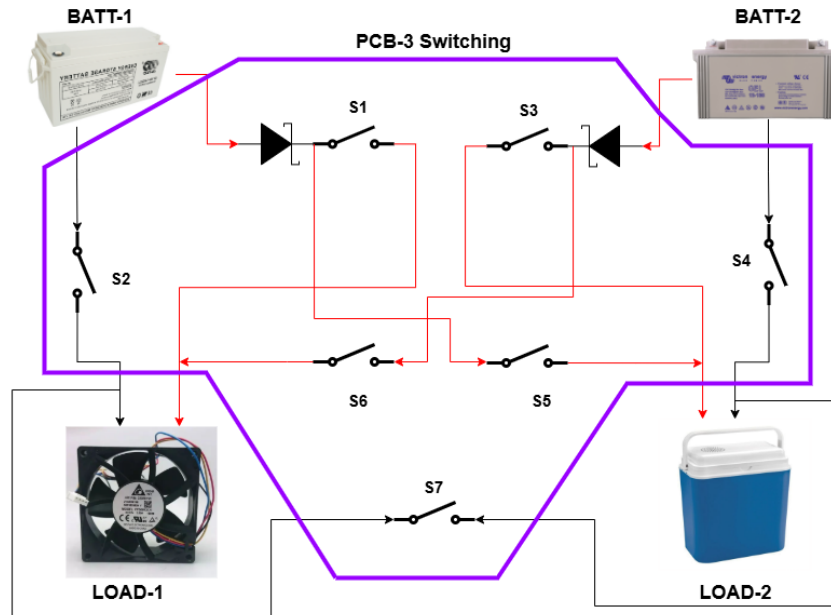


Figure 3.19: PCB3 Arcitecture for switching circuit .

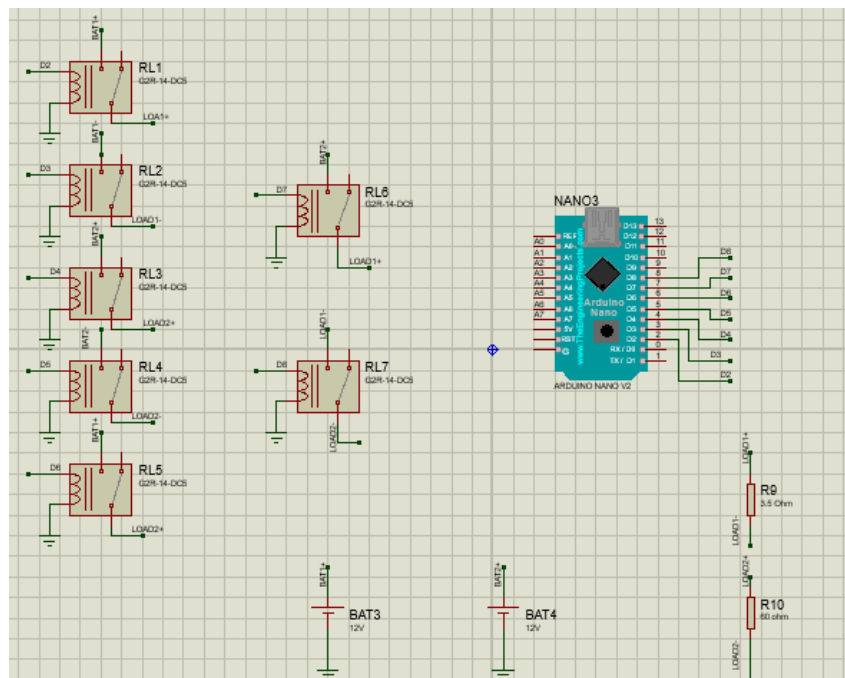


Figure 3.20: PCB3 schematic for switching circuit .

Components: Arduino Uno and a relay module with seven electromechanical relays.

Function: PCB3 manages the switching of power paths between the two microgrids (MG1 and MG2), the storage batteries, and the loads. Each relay acts as an **Automatic**

Transfer Switch (ATS) element, enabling or disabling connections according to the control logic received from the higher-level controller.

Relay Specifications: The relays used are rated for:

- Coil voltage: **5V DC** (driven by Arduino Uno GPIO pins through a driver stage).
- Contact rating: **10A at 250V AC / 30V DC**, ensuring safe operation for typical microgrid-scale loads.
- Switching delay: approximately **10–15 ms**, which is acceptable for low-voltage DC microgrid applications.

Control Logic: The Arduino Uno receives a serial command in the form of a single number (1, 2, or 3). Each value corresponds to a predefined operating mode:

- **Mode 1:** Each battery supplies only its respective load (Battery 1 $\rightarrow$ Load 1, Battery 2 $\rightarrow$ Load 2).
- **Mode 2:** Assistance mode where Battery 1 supplies both Load 1 and Load 2.
- **Mode 3:** Assistance mode where Battery 2 supplies both Load 1 and Load 2.

Based on the received command, the Arduino activates or deactivates the corresponding relays, ensuring safe switching between modes.

3.3 Software Design

This section describes the software architecture developed to manage data acquisition, control, switching, and supervision of the dual microgrid system. The design is organized in four main layers: embedded software for measurement and actuation, Python scripts for computation and decision-making, Node-RED flows for communication and orchestration, and a ThingsBoard dashboard for remote supervision.

3.3.1 Embedded Software (Arduino)

The Arduino boards (Nano for measurement PCBs, Uno for switching PCB) were programmed with lightweight firmware to ensure reliable operation.

Firmware Logic:

- Acquisition of sensor data: voltage and current from PV2, Battery 1, Battery 2, and relays status.

- Serial communication with the Raspberry Pi for real-time data transfer.
- Relay activation according to control commands (Mode 1, Mode 2, Mode 3).

Sampling Rate and Resolution: Measurements are sampled at a fixed rate of 1 Hz (once per second). The Arduino ADC provides a 10-bit resolution, which, combined with the chosen voltage divider ratios and ACS712 sensors, ensures sufficient accuracy for the system's voltage/current ranges.

Limitations: Due to the limited processing capacity of Arduino boards, only basic data acquisition and relay control are implemented, while higher-level decision-making is delegated to the Raspberry Pi.

3.3.2 Raspberry Pi Scripts (Python)

The Raspberry Pi serves as the central processing unit. A set of Python scripts perform the following tasks:

- **SOC Estimation:** State of charge of both batteries is computed using coulomb counting and power balance equations.
- **Energy Balance:** Daily and cumulative energy production (PV1, PV2, wind) and consumption (Load1, Load2) are calculated.
- **Switching Decision:** The algorithm selects the operating mode (1, 2, or 3) based on SOC thresholds and the Auto/Manual state received from ThingsBoard.
- **Data Formatting:** All processed data are structured into JSON objects before being sent to Node-RED.

This layer also integrates weather data (solar irradiation, wind speed and direction) to validate correlations between environmental conditions and generation output.

3.3.3 Node-RED Flows

Node-RED acts as the middleware orchestrating data flows between devices and platforms.

Main Functions:

- Data acquisition from Arduino boards (measurement and switching PCBs).
- Bluetooth communication with the secondary Raspberry Pi to retrieve PV1 and wind turbine measurements.

- Bidirectional communication with the Python script (input: raw measurements, output: processed SOC and decisions).
- Forwarding of all data to ThingsBoard via MQTT/HTTP.
- Local supervision through interactive flows for monitoring and debugging.

Mode Control: When the operator selects Auto/Manual or Mode 1/2/3 on ThingsBoard, Node-RED ensures synchronization with the Python script and relays the command to the Arduino Uno responsible for switching.

3.3.4 ThingsBoard Supervision

The ThingsBoard platform provides a remote supervision interface.

Dashboard Design:

- Real-time visualization of PV, wind, and battery parameters.
- SOC display with graphical indicators.
- Current operating mode and recommended mode based on SOC algorithm.
- Energy production/consumption history through charts.

Remote Control:

- Buttons to toggle between Auto/Manual control.
- Commands to select Mode 1, Mode 2, or Mode 3.
- Confirmation feedback to ensure reliability of switching operations.

Data Storage and Visualization: All historical data are logged in ThingsBoard for further analysis. This includes battery cycles, renewable generation trends, and load profiles, which are essential for validation in the testing phase.

3.4 Communication Architecture

The communication architecture ensures seamless data exchange between the measurement circuits, control units, supervision interfaces, and the cloud platform. It is structured into two levels: **internal communication** (within the local system) and **external communication** (with remote supervision).

Internal Communication

- **Arduino → Raspberry Pi:** Serial communication is used to transmit raw data from the measurement circuits (PCB1 and PCB2) and to receive switching commands from the central controller.
- **Raspberry Pi → Arduino (PCB3):** Switching commands are transmitted via serial interface. The Arduino Uno interprets numeric codes (1, 2, or 3) and actuates relays accordingly.
- **Bluetooth Link:** A secondary Raspberry Pi acquires wind turbine and PV1 measurements from the hybrid controller via Bluetooth. These values are then forwarded to the central Raspberry Pi using MQTT messages.
- **Weather Station:** The weather station Arduino collects environmental parameters such as solar irradiation, wind speed and orientation, temperature, and humidity. These values are first sent to a dedicated Raspberry Pi, which forwards them to the central Raspberry Pi via MQTT..
- **Raspberry Pi → Node-RED → Python:** Node-RED collects raw measurements and forwards them to Python scripts for processing (SOC estimation, energy balance, switching decision). The results are then sent back to Node-RED for visualization and forwarding.

External Communication

- **Node-RED → ThingsBoard:** Data is published to the ThingsBoard cloud using MQTT and HTTP protocols. This allows real-time visualization of measurements, SOC values, and switching states.
- **ThingsBoard → Node-RED:** Remote control commands (Auto/Manual, Mode 1/2/3) are received from the ThingsBoard dashboard and transmitted back into the system through Node-RED, which communicates with Python scripts and Arduino units.

Communication Flow Diagrams

The communication layers are illustrated through two complementary figures.

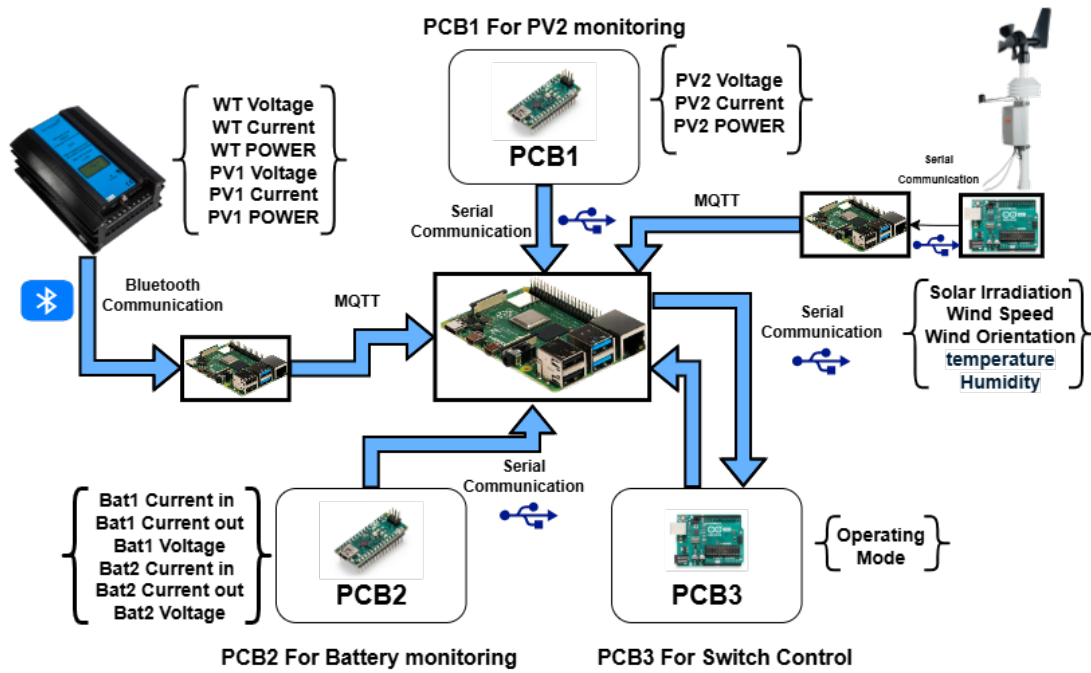


Figure 3.21: Communication architecture – Data acquisition layer .

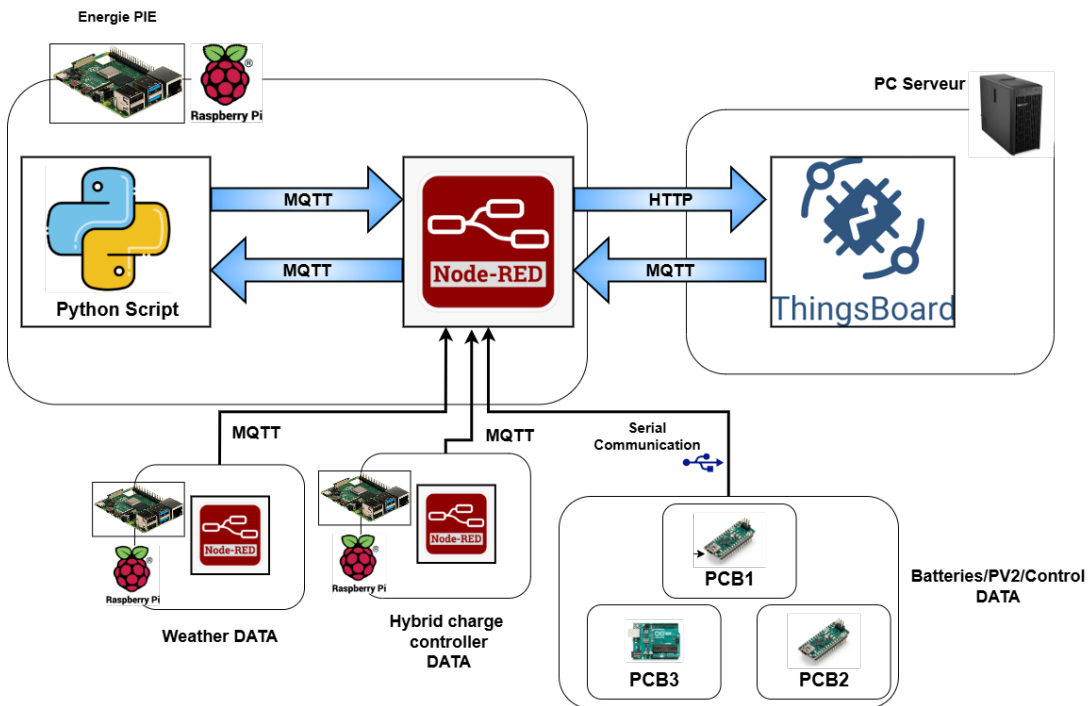


Figure 3.22: Communication architecture – Data processing and supervision layer .

3.5 Control and Switching Logic

3.5.1 Operation Modes

The system operates under three distinct modes depending on the state-of-charge (SOC) of the two batteries:

- **Mode 1 (Normal Operation):** Battery 1 supplies Load 1 and Battery 2 supplies Load 2 independently. This is the default configuration when both batteries are in a healthy SOC range.
- **Mode 2 (Energy Sharing MG1 $\rightarrow$ MG2):** Battery 1 assists Battery 2 in supplying Load 2 when Battery 2 is undercharged.
- **Mode 3 (Energy Sharing MG2 $\rightarrow$ MG1):** Battery 2 assists Battery 1 in supplying Load 1 when Battery 1 is undercharged.

Both automatic and manual modes are available. In automatic mode, the switching is determined by the SOC-based algorithm. In manual mode, the operating mode can be selected via the ThingsBoard interface.

3.5.2 SOC-Based Switching Algorithm

To avoid rapid oscillations due to small SOC fluctuations, a bidirectional hysteresis mechanism is applied. The algorithm uses the following thresholds:

- **High threshold (S_{haut}):** 80% SOC.
- **Low threshold (S_{bas}):** $S_{haut} - h = 77\%$, where $h = 3\%$ is the hysteresis margin.
- **Activation threshold (S_{act}):** $S_{haut} + h = 83\%$.

The decision logic can be summarized as follows:

1. If Battery 1 SOC $< S_{bas}$ and Battery 2 SOC $\geq S_{haut}$, switch to **Mode 3**.
2. If Battery 2 SOC $< S_{bas}$ and Battery 1 SOC $\geq S_{haut}$, switch to **Mode 2**.
3. If both batteries are above or both below S_{haut} , remain in **Mode 1**.
4. If none of the above conditions are met, retain the previously recommended mode.

3.5.3 Flowchart of Decision Logic

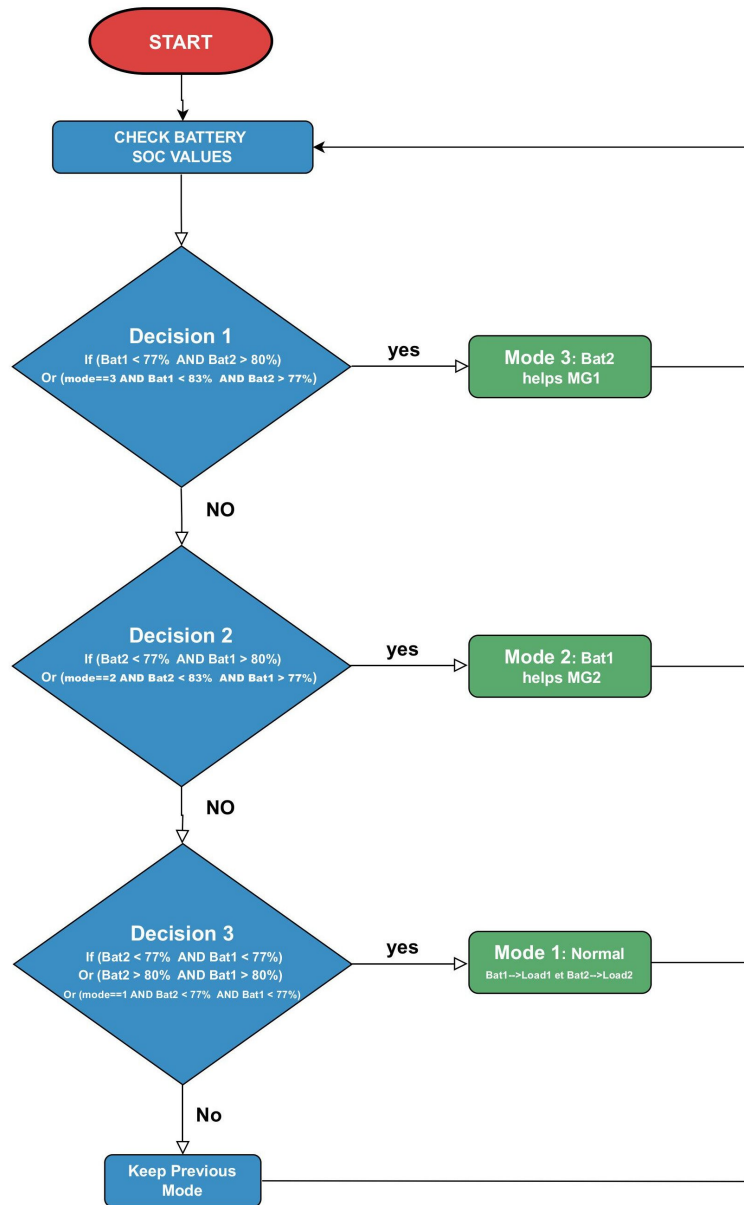


Figure 3.23: Flowchart of SOC-based switching algorithm with hysteresis.

3.6 System Diagrams and Illustrations

To better understand the proposed solution, the following diagram presents the global architecture of the system. It illustrates the integration of renewable energy sources (wind and photovoltaic), storage units, monitoring modules, and the switching mechanism for load management. This global view highlights the interconnection between acquisition, processing, and decision-making layers.

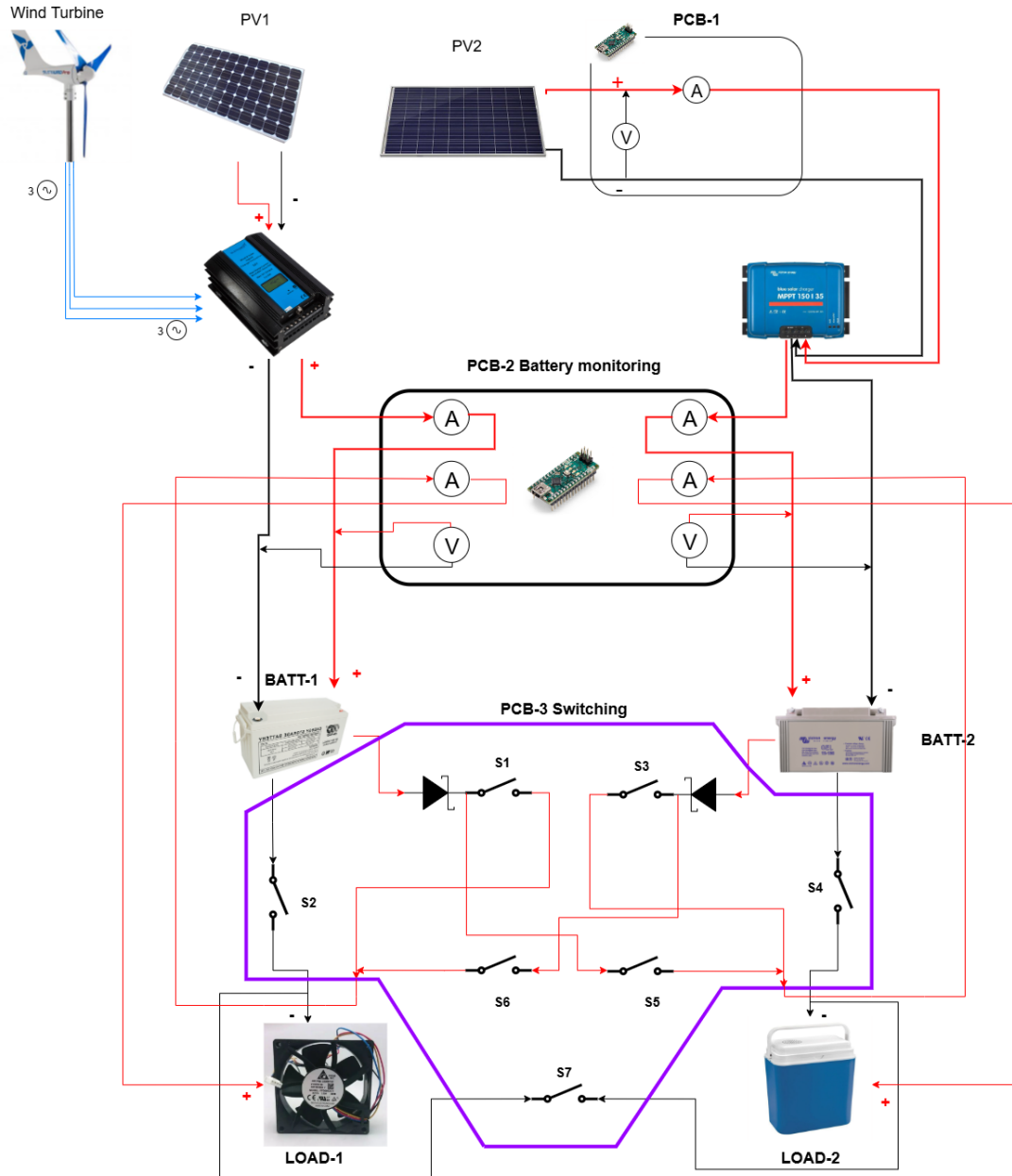


Figure 3.24: Global architecture of the proposed system, showing energy sources, batteries, monitoring boards, and switching module.

Conclusion

In this chapter, the system architecture and its functional blocks were presented, including the data acquisition, monitoring, switching, and communication layers. These diagrams and descriptions establish a clear foundation for the next stage. The following chapter will focus on the **implementation**, where the designed architecture is translated into practical hardware and software realizations.

Chapter 4

Implementation and Validation

Introduction

Following the system design phase, this chapter focuses on the practical realization of the proposed solution. The implementation is divided into two main parts: hardware prototyping and software integration. The hardware part includes the development of dedicated PCBs, the switching module, and the central Raspberry Pi unit. The software part covers data acquisition, SOC-based control logic, and supervisory communication through Node-RED and ThingsBoard. Together, these elements form a fully functional platform ready for testing and performance evaluation in the next chapter.

4.1 Hardware Implementation

This section presents the practical realization of the hardware elements designed in the previous chapter. It covers the fabrication of the dedicated printed circuit boards (PCBs), the integration of external measurement devices, and the centralization of all signals within the Raspberry Pi for subsequent processing and control. Photographs and diagrams of the developed hardware are provided to illustrate the implementation.

4.2 Hardware Implementation

This section presents the practical realization of the hardware elements designed in the previous chapter. It covers the fabrication of the dedicated printed circuit boards (PCBs), the integration of external measurement devices, and the centralization of all signals within the Raspberry Pi for subsequent processing and control.

4.2.1 PCB-1: PV2 Measurement Board

The first printed circuit board (PCB-1) is dedicated to monitoring the electrical parameters of the second PV source **PV2**. It integrates an **Arduino** microcontroller, a voltage divider to step down the DC bus voltage, and an **ACS712** current sensor for current measurement. The Arduino samples these analog signals, converts them into digital values, and transmits them via serial communication to the central Raspberry Pi unit.

During prototyping, particular attention was given to the design of the voltage divider (resistor ratio ensuring measurement up to 36 V within the Arduino's 5V ADC range) and to the output filtering of the ACS712 to reduce noise. The board was fully designed, soldered, and tested to ensure stable operation.

Result example (Arduino JSON output):

```
{"PV2_voltage":32,"PV2_current":6.5,"PV2_power":208}
```

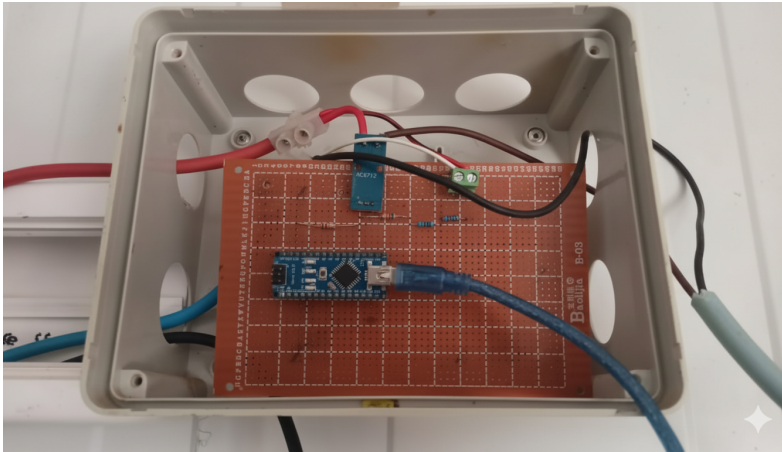


Figure 4.1: Prototype of PCB-1

4.2.2 PCB-2: Battery Monitoring Board

The second printed circuit board **PCB-2** is responsible for monitoring the electrical behavior of both batteries (Bat1 and Bat2) in the dual-microgrid system. It integrates:

- Four **ACS712** sensors for measuring the input and output currents of each battery,
- Two voltage dividers for capturing the terminal voltages of the batteries,
- An Arduino microcontroller for data acquisition and transmission.

These measurements are critical for estimating the state of charge (SOC), detecting charging/discharging conditions, and monitoring the energy flows between the microgrids. The PCB was designed with dedicated connectors to ensure reliable attachment to each battery branch, and the Arduino was programmed to continuously acquire and forward the data to the Raspberry Pi.

Result example (Arduino JSON output):

```
{"bat1_voltage":12.75,"bat1_in":0,"bat1_out":0.167,  
  "bat2_voltage":12.65,"bat2_in":0,"bat2_out":3.467}
```

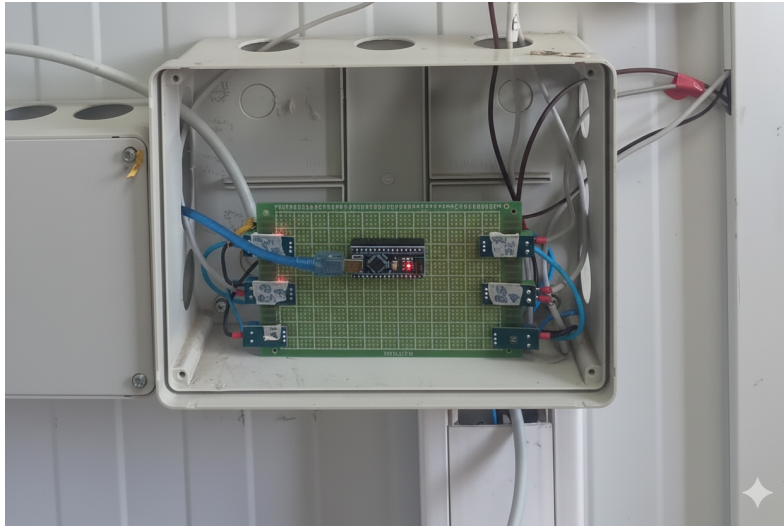


Figure 4.2: Prototype of PCB-2.

4.2.3 PCB-3: Switching Module

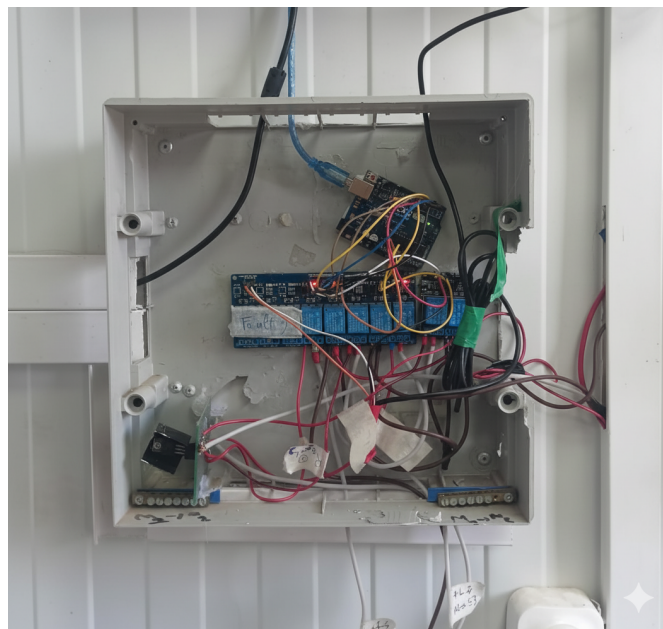


Figure 4.3: Prototype of PCB-3.

The third printed circuit board (PCB-3) implements the switching logic of the microgrid system. It consists of an **Arduino** Uno controlling seven mechanical relays, each connected to a specific battery or load line. This module executes the switching commands

received from the Raspberry Pi via serial communication. By energizing the appropriate relay channels, the Arduino dynamically reallocates the loads between the two microgrids according to the operating mode (Mode 1, Mode 2, or Mode 3).

For flexibility, a manual override mechanism was also implemented, allowing the operator to test or debug the switching logic independently of the automatic control.

Result example (Arduino JSON output):

```
{"status":"active","mode":1,"message":"Mode 1 actif","timestamp":978684451}
```

4.2.4 Integration of External Measurements

In addition to the custom PCBs, further measurements are integrated from external sources:

- The first Raspberry Pi (connected via Bluetooth to the hybrid PV/wind controller) acquires the voltage and current of PV1 and the wind turbine.
- A professional weather station was deployed to provide environmental data. It combines:
 - An **ABB** station with a wind vane for wind orientation,
 - A Kipp & Zonen pyranometer (0–1 V output) for solar irradiation,
 - A **DHT22** sensor for temperature and humidity.

These measurements are fused with the electrical data to enable correlation analyses between resource availability (solar irradiance, wind speed) and energy production.



Figure 4.4: Weather station and external sensors

4.2.5 Raspberry Pi Central Unit

All PCBs and external devices are connected to a central Raspberry Pi, which serves as the core of the hardware implementation. The Raspberry Pi aggregates the data coming from PCB-1, PCB-2, and PCB-3, as well as the Bluetooth link with the hybrid controller. It runs Python scripts to compute derived parameters such as SOC, instantaneous powers, and cumulative energy balances. It also applies the decision-making logic for switching between modes, based on battery SOC thresholds.

This centralization ensures synchronized operation of the entire system, while preparing the processed data for Node-RED and cloud supervision via ThingsBoard.

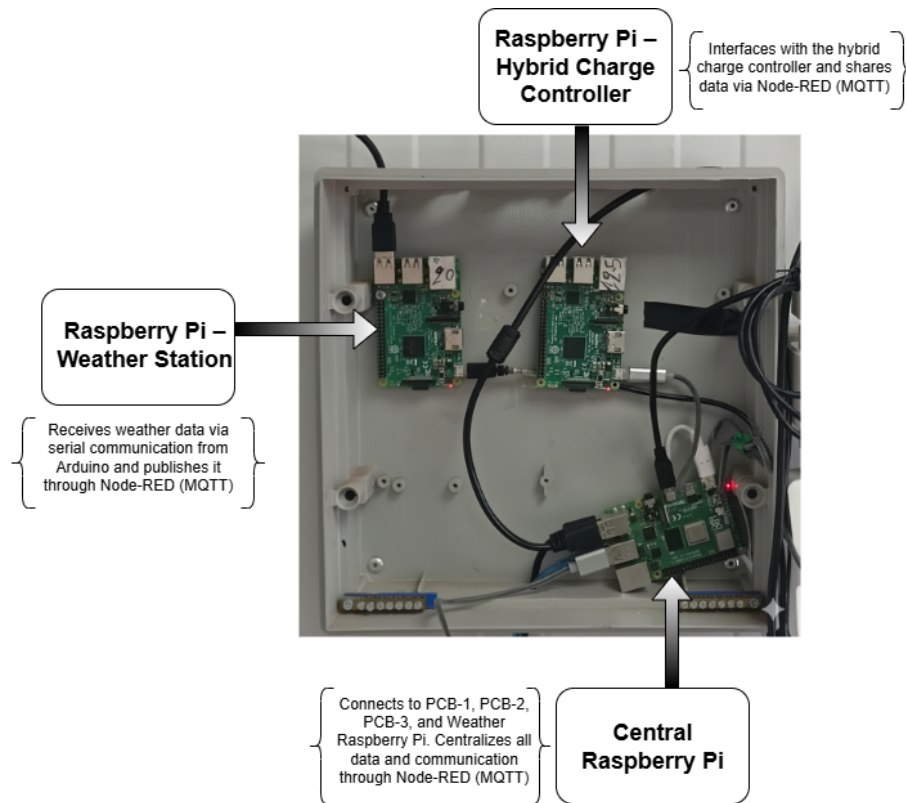


Figure 4.5: Raspberry Pi central unit connected to PCBs and Bluetooth receiver for data aggregation and processing.

4.3 Software Implementation

The software implementation ensures the integration between hardware measurements, data processing, and supervisory communication. It relies on two core components deployed on the central **Raspberry Pi**:

- **Node-RED**, responsible for data acquisition, pre-processing, communication with external platforms, and saving results in CSV logs,

- **Python scripts**, dedicated to advanced computations such as SOC estimation, energy balance, diagnostics, and decision-making for the switching algorithm.

4.3.1 Data Acquisition Layer

The data acquisition is orchestrated through Node-RED, which collects:

- Serial inputs from the Arduino-based PCBs (PCB-1, PCB-2, PCB-3),
- MQTT messages from the secondary Raspberry Pi connected to the hybrid PV/Wind charge controller,
- Weather data from the ABB station, pyranometer, wind vane, and DHT22 (two wind orientation sensors combined for higher reliability).

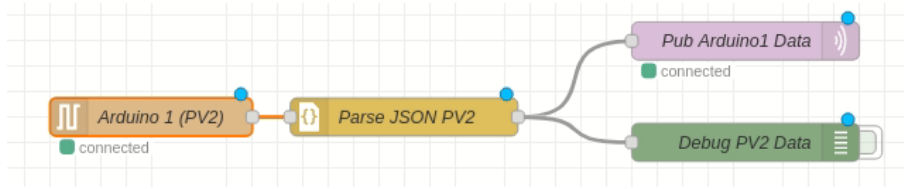


Figure 4.6: Node-RED flow for data acquisition from PCB-1 (PV2 board).

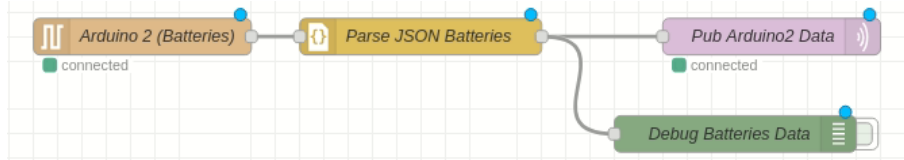


Figure 4.7: Node-RED flow for data acquisition from PCB-2 (Baterries board).

Node-RED is also responsible for saving the processed results into CSV files every 5 minutes. When the Raspberry Pi restarts, the flow reloads the last values from the CSV file to resume SOC and energy accumulation without loss of continuity.

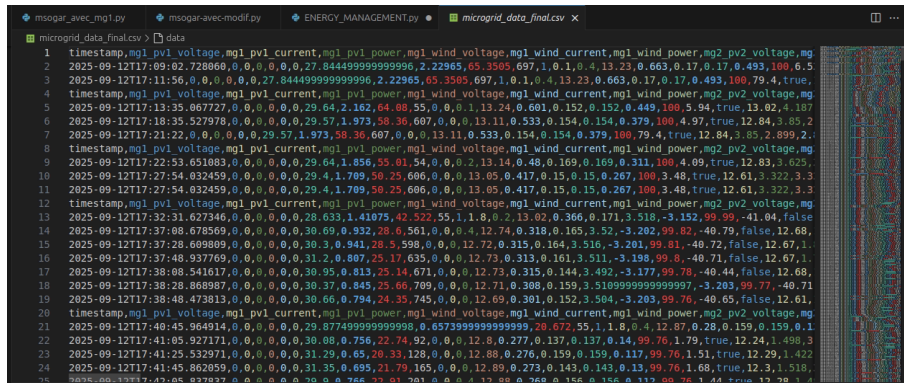


Figure 4.8: Example of CSV log file generated by Node-RED

4.3.2 Python Processing Layer

Python scripts run in real time, directly connected to Node-RED through bidirectional communication. Their main tasks include:

- Estimation of battery SOC using a **hybrid algorithm**,
- Computation of instantaneous power and cumulative energies (daily and total),
- Diagnostic checks for sensor reliability and message integrity.

Hybrid SOC estimation algorithm The SOC estimation combines **Coulomb Counting** when significant current is present, and a **voltage-based method** when the battery is in a near-idle state for more than 5 minutes.

Listing 4.1: SOC calculation using Coulomb Counting

```

1  def calculer_soc_coulomb_counting(self, soc_actuel, courant_net
    , voltage, capacite, dt):
2  """Calcule le SOC par la méthode Coulomb Counting"""
3  efficacite = 0.95 if courant_net > 0 else 1.0
4
5  delta_ah = (courant_net * dt / 3600) * efficacite
6  delta_soc = (delta_ah / capacite) * 100
7
8  nouveau_soc = soc_actuel + delta_soc
9  return max(0, min(100, nouveau_soc))

```

Listing 4.2: SOC estimation based on voltage thresholds

```

1  def calculer_soc_tension(self, voltage, capacite=None):
2  """Calcule le SOC basé sur la tension"""
3  courbe = [
4  (11.8, 20),
5  (12.0, 40),
6  (12.2, 60),
7  (12.4, 80),
8  (12.6, 100)
9  ]
10
11  if voltage <= 11.8:
12  return 10
13  if voltage >= 12.6:
14  return 100
15

```

```

16     for i in range(len(courbe) - 1):
17         v1, soc1 = courbe[i]
18         v2, soc2 = courbe[i + 1]
19
20         if v1 <= voltage <= v2:
21             soc = soc1 + (voltage - v1) * (soc2 - soc1) / (v2 - v1)
22             return round(soc, 1)
23
24     return voltage * 10

```

4.3.3 Communication Layer

Three levels of communication are implemented:

- **Node-RED - Python:** real-time bidirectional exchange, Python receiving raw data and returning processed JSON results.
- **Node-RED - Arduino:** serial communication for sending mode and switching commands to the relay board.
- **Node-RED - ThingsBoard:** HTTP is used for sending telemetry in real time, while MQTT is reserved for receiving configuration updates and manual/automatic control states.

4.3.4 SOC-Based Switching Algorithm

The final step of the processing chain is the SOC-based switching algorithm. It applies hysteresis thresholds to avoid oscillations and determines whether the system should operate in Mode 1 (normal), Mode 2 (Bat1 assists MG2), or Mode 3 (Bat2 assists MG1).

Listing 4.3: SOC-based switching algorithm with hysteresis

```

1     def determiner_mode(self):
2         """Détermine le mode de fonctionnement basé sur les SOC avec
3             hystérésis"""
4
5         hystere = 3
6         seuil_haut = 80
7         seuil_bas = seuil_haut - hystere
8         seuil_activation = seuil_haut + hystere
9
10        # Mode 3: Bat2 aide MG1
11        if (self.bat1_soc < seuil_bas and self.bat2_soc >= seuil_haut)
12            or \

```

```

11     (self.mode_recommande == 3 and self.bat1_soc < seuil_activation
12         and self.bat2_soc >= seuil_bas):
13
14     # Mode 2: Bat1 aide MG2
15     elif (self.bat2_soc < seuil_bas and self.bat1_soc >= seuil_haut
16         ) or \
17     (self.mode_recommande == 2 and self.bat2_soc < seuil_activation
18         and self.bat1_soc >= seuil_bas):
19     return 2
20
21     # Mode 1: Mode normal
22     elif (self.bat1_soc >= seuil_haut and self.bat2_soc >=
23         seuil_haut) or \
24     (self.bat1_soc < seuil_haut and self.bat2_soc < seuil_haut) or
25     \
26     (self.mode_recommande == 1 and self.bat1_soc >= seuil_bas and
27         self.bat2_soc >= seuil_bas):
28     return 1
29
30     else:
31     return self.mode_recommande

```

4.3.5 Processed Results

After computation, Python sends the processed results back to Node-RED in JSON format, which are then forwarded to supervision or stored in CSV.

Listing 4.4: Example of JSON results received from Python

```

{
    "timestamp": "2025-09-28T00:58:09.906483",
    "microgrid1": {"pv1": {"voltage": 0, "current": 0, "power": 0}},
    "microgrid2": {"pv2": {"voltage": 0.87, "current": -0.044, "
        power": -0.04}},
    "batteries": {
        "bat1": {"voltage": 12.73, "soc": 99.23, "power_net":
            -1.96},
        "bat2": {"voltage": 12.61, "soc": 91.08, "power_net":
            -43.64}
    },
    "loads": {"charge1_power": 1.96, "charge2_power": 43.64},
    "operation": {"mode_recommande": 1, "mode_confirme": 1, "
        mode_description": "Mode normal"}
}

```

```
}

```

This structured JSON enables seamless integration with the supervision dashboard and facilitates offline data analysis.

4.4 Dashboard and Supervision (ThingsBoard)

The supervision and monitoring of the microgrid system are performed through **ThingsBoard**, an open-source IoT platform that provides data visualization, device management, and bidirectional communication. This platform plays a central role in ensuring both real-time monitoring and control capabilities.

4.4.1 Data Transmission and Command Flow

Data exchange between the Raspberry Pi and ThingsBoard is managed through **Node-RED**. The process follows three complementary paths:

- **Telemetry upload:** At each cycle, Node-RED collects all measurements (voltages, currents, power, SOC, energy balance, etc.) and sends them in a single JSON object via HTTP to ThingsBoard.

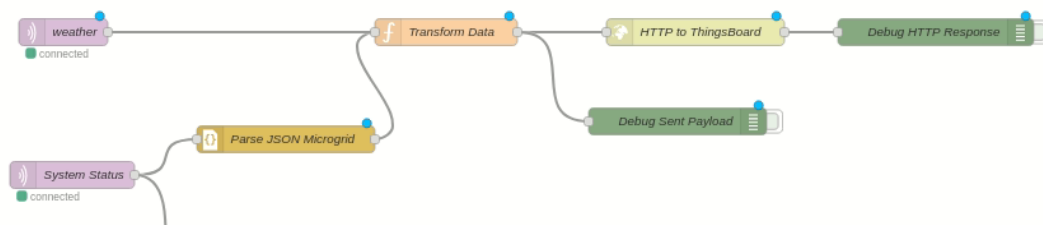


Figure 4.9: Node-RED flow for sending telemetry data to ThingsBoard via HTTP.

- **Command reception (RPC):** When a mode change or control command is triggered from ThingsBoard, it is sent through the MQTT topic: `v1/devices/me/rpc/request/+`. Node-RED receives and parses the RPC message, then forwards it to the Python processing layer, which is solely responsible for executing the requested action.

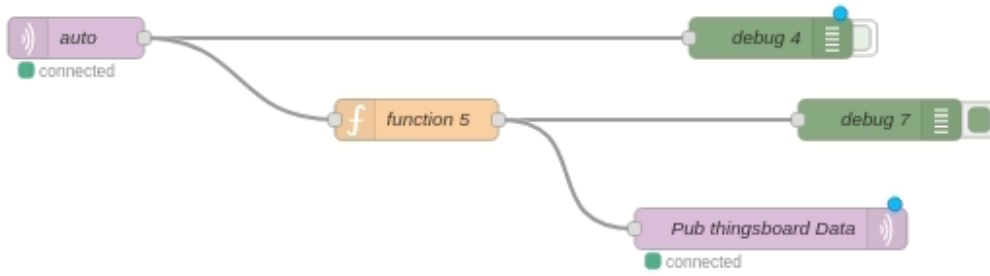


Figure 4.10: Node-RED flow for receiving and parsing RPC commands from ThingsBoard.

- **Attribute synchronization:** Attributes (`v1/devices/me/attributes`) are used to synchronize the dashboard widgets with the real system state. For example, when the system is in manual mode (decided by the Python script), the corresponding button on the ThingsBoard dashboard automatically updates to “Manual” to reflect the actual mode.

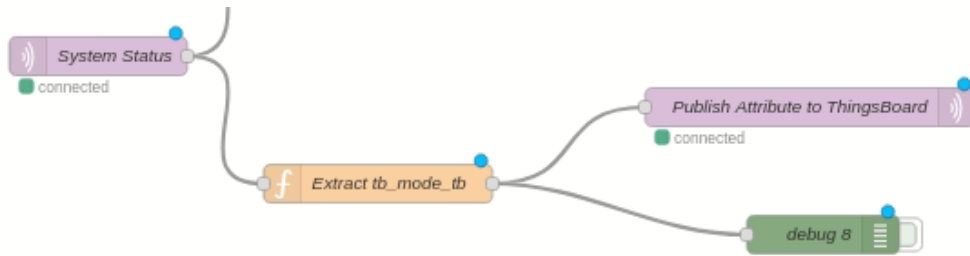


Figure 4.11: Node-RED flow to sync ThingsBoard attributes with real system state.

4.4.2 Dashboard Structure

The dashboard was designed as a **multi-state interface**, where each state corresponds to a specific perspective of the system:

- **Home:** General description of the project with a navigation button to enter the global dashboard.
- **Global Dashboard:** Overview of both microgrids, including SOC of batteries, daily energy production, load consumption, net battery power, energy balance, and a central card summarizing total energy production from PV1, PV2, wind turbine, and loads.
- **Batteries:** Dedicated to battery monitoring. Left and right sections show SOC, voltage, net current, net power, cumulative energy, and charging/discharging status for each battery. The central card illustrates SOC evolution over time.

- **Microgrid 1:** Presents PV1 voltage/current/power, wind turbine voltage/current/power, load consumption, shared energy to MG2, cumulative energy, and a daily evolution card showing production, consumption, and balance.
- **Microgrid 2:** Similar to MG1 but without wind turbine data (since MG2 does not have one).
- **Mode & Control:** Shows the current operating mode, recommended and confirmed modes, synchronization status, and manual/automatic control. Includes switches for Auto/Manual and buttons to select one of the three modes in manual mode.
- **Flow and Transfer:** Displays total and daily shared energy between microgrids, along with a card showing switching mode over time.
- **History:** A data table of all system parameters. The user can select a time period to visualize past data. Data storage is handled automatically by the Cassandra database, while advanced queries can be executed via the ThingsBoard REST API.

4.4.3 ThingsBoard Device and Alias Configuration

On the ThingsBoard side, the device configuration defines telemetry keys, attribute mapping, and dashboard entities. Entity aliases are used to dynamically link data sources to widgets, ensuring flexibility and reusability.

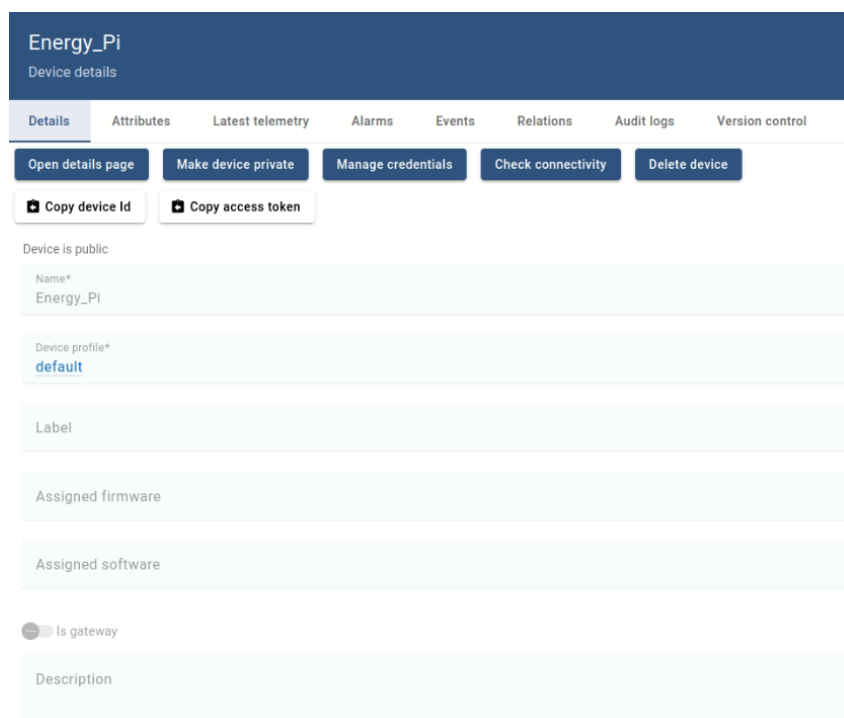


Figure 4.12: Device configuration in ThingsBoard for microgrid supervision.

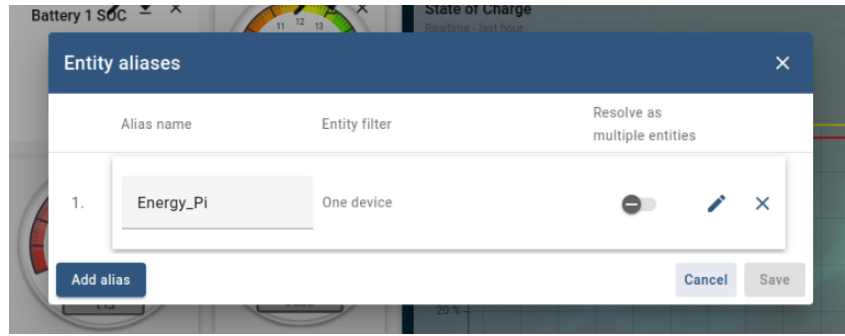


Figure 4.13: Entity alias configuration linking telemetry to dashboard widgets.

4.4.4 Access and User Roles

To ensure system integrity, dashboard access is restricted to laboratory members as **viewers**, without editing rights. This guarantees reliable supervision without risk of unintentional modifications.

4.4.5 Supervision Windows and Interface Snapshots

To illustrate the developed supervision interface, several windows of the ThingsBoard dashboard are presented. Figures A.2, A.3, and A.6 highlight the most representative states of the platform: the global overview, the battery-focused view, and the mode and control panel.



Figure 4.14: Global dashboard: microgrids overview, battery SOC, daily production, load consumption.



Figure 4.15: Battery dashboard: SOC, voltage, net power, energy, charge/discharge.



Figure 4.16: Modes & control dashboard: current/recommended mode, auto/manual switch, manual controls..

For the sake of readability, the remaining dashboard states (*Home*, *Microgrid 1*, *Microgrid 2*, *Flow and Transfer*, and *History*) are provided in Annex A.

4.5 Experimental Validation

4.5.1 Global System Setup

The experimental setup integrates the main components of the proposed microgrid test-bench, including the PV modules with MPPT controllers, battery banks (Battery 1 (Batt1) and Battery 2 (Batt2)), loads (Load of Microgrid 1 (Load1) and Load of Microgrid 2 (Load2)), the hybrid charge controller, and the designed electronic boards placed inside protective boxes. At the center of the installation, the supervision dashboard provides real-time monitoring of system states and energy flows. Figure 4.17 illustrates the overall laboratory setup.

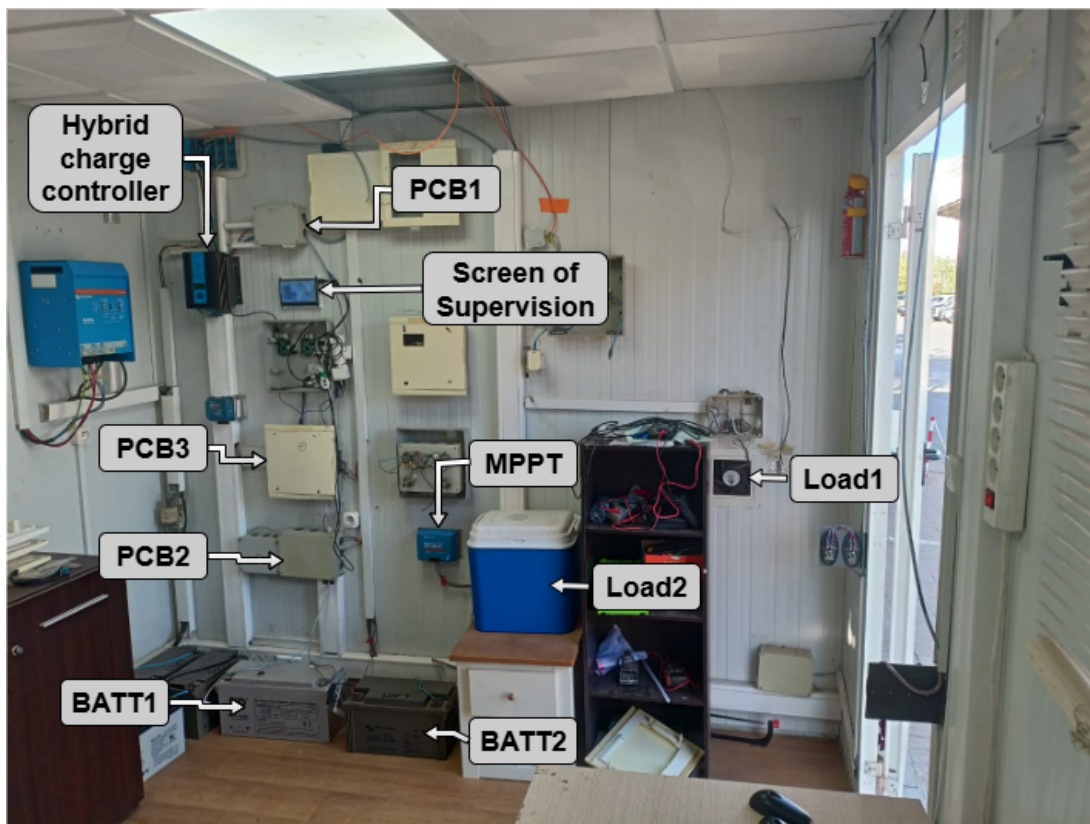


Figure 4.17: Global experimental setup .

4.5.2 Validation of PV Production

The validation of the PV subsystem was carried out by comparing the measured production with the irradiation data collected from the pyranometer. In Figure 4.18, we present the validation of cumulative energy production of PV_1 compared to solar irradiation. Additional validation results, including the energy production of PV_2 as well as the instantaneous power of PV_1 and PV_2 with respect to irradiation, are reported in Annex ??.

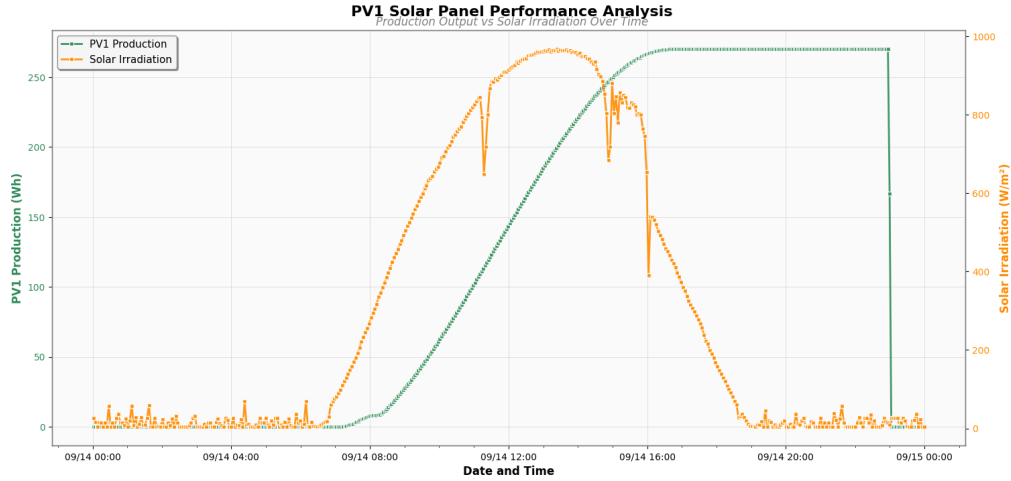


Figure 4.18: Validation of cumulative energy production of PV_1 with respect to solar irradiation.

4.5.3 Validation of Wind Turbine Production

To validate the wind energy subsystem, the measured wind turbine power was compared qualitatively to the wind speed profile. As depicted in Figure B.5, the power output follows the same trend as the wind speed, confirming the proper functionality of the conversion chain. However, due to the low wind speed at the installation site (rarely exceeding 2 m/s), the overall production remained limited.

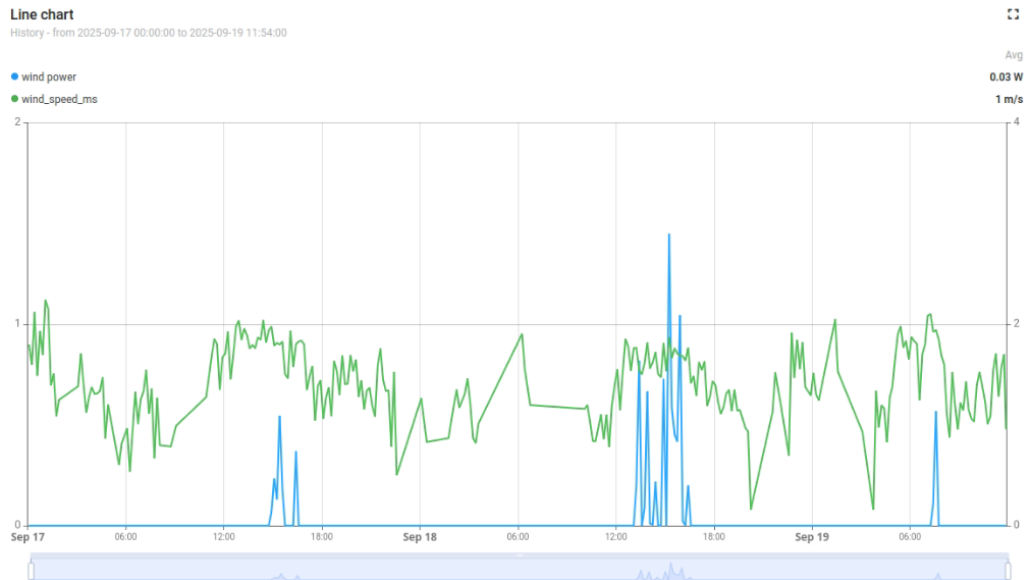


Figure 4.19: Qualitative validation of wind turbine production: power versus wind speed.

4.5.4 Validation of Interconnection and SOC-Based Switching

The interconnection between the two microgrids was tested under different battery states of charge (SOC). Figure 4.20 shows the transitions between the operating modes (Mode 1, Mode 2, and Mode 3), highlighting the efficiency of the automatic switching strategy. The results confirm that the loads are correctly transferred between batteries when the threshold of 30% SOC is reached.

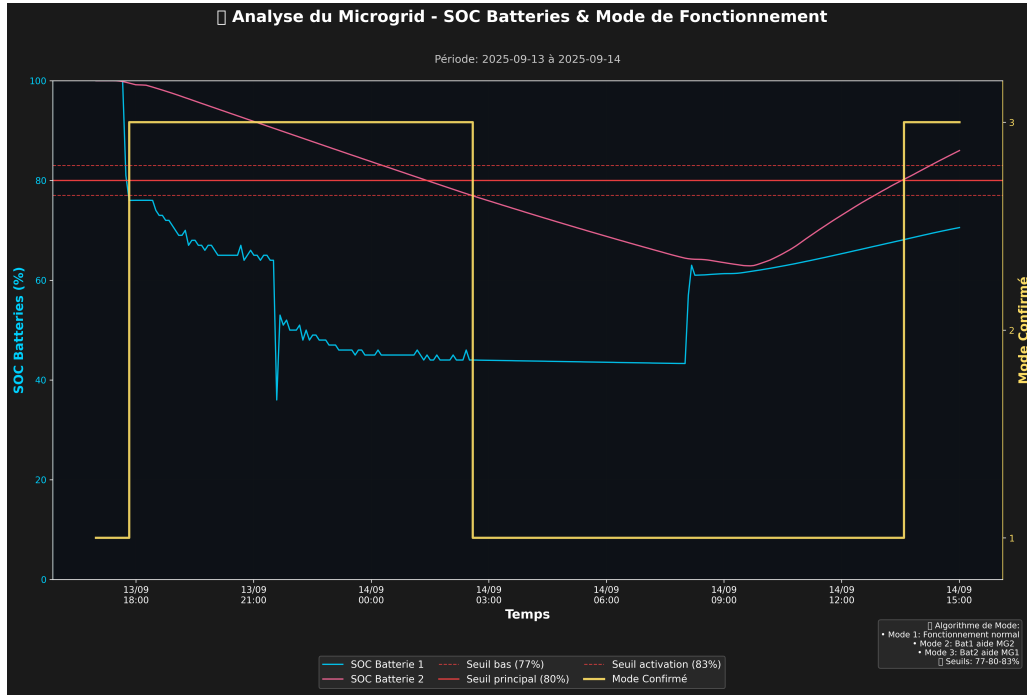


Figure 4.20: Experimental validation of interconnection and SOC-based switching between the two microgrids.

4.5.5 Discussion of Results

The obtained results confirm the correct operation of the global system. The PV production matches well with the irradiation profile, validating the effectiveness of the implemented MPPT control. The wind turbine subsystem demonstrates coherence between wind speed and produced power, though the low wind speed at the installation site strongly limited its contribution to the total energy balance. Regarding the interconnection, the SOC-based switching strategy ensured reliable and autonomous transfer between the two batteries.

Some limitations were identified during the experimental campaign. In particular, the voltage dividers used for battery measurement relied on resistors with non-ideal tolerances, leading to slight variations in the estimated battery voltage and, consequently, in the computed SOC. Despite these limitations, the system has been fully validated and demonstrated robustness in all tested scenarios.

Conclusion

In this chapter, the proposed system was implemented and validated. The hardware and software integration proved effective, and the experimental results confirmed the correct behavior of PV, wind, and switch-based control strategies.

General Conclusion

The transition to sustainable and decentralized energy systems requires innovative solutions capable of integrating renewable sources while ensuring reliability and efficiency. This work has addressed these challenges through the development of a comprehensive microgrid monitoring and control platform, combining renewable generation modeling, advanced SOC estimation, embedded data acquisition, and supervisory control integration. The proposed system demonstrates the feasibility of unifying hardware and software layers to achieve real-time monitoring, intelligent decision-making, and reliable energy management.

The main contributions of this work can be summarized as follows:

- Modeling and analysis of renewable energy sources (PV and wind) and their integration within a microgrid environment.
- Implementation of advanced SOC estimation techniques for accurate battery monitoring.
- Design and realization of embedded acquisition systems for real-time data collection and processing.
- Development of a supervisory control architecture using IoT-based solutions such as Node-RED and ThingsBoard.
- Proposal of intelligent switching and load management strategies to enhance operational reliability.

Despite these achievements, certain limitations remain, which open several promising perspectives for future research and development:

- **Integration of weather prediction into switching algorithms:** incorporating short- and medium-term forecasts (solar irradiance, wind speed) to anticipate production deficits and optimize energy allocation decisions.
- **Advanced SOC estimation techniques:** exploring the use of artificial intelligence models and advanced filters such as Kalman filtering to improve accuracy and robustness under dynamic operating conditions.

- **Real-time adaptive AI models:** developing machine learning models capable of continuous training with live data to predict future consumption patterns and integrate them into the supervisory switching strategies.
- **Blockchain-based energy transaction management:** extending the system toward decentralized energy trading and automated billing between interconnected microgrids, ensuring transparency and security of energy exchanges.

In conclusion, this work contributes to the ongoing efforts toward smart and sustainable energy systems. By combining renewable integration, advanced monitoring, and intelligent supervisory control, it highlights the importance of multidisciplinary approaches in addressing the technical and environmental challenges of the energy transition. The perspectives outlined above pave the way for further innovations in microgrid design and operation, reinforcing their role as key enablers of a greener and more resilient energy future.

Appendix A

Dashboard Windows (ThingsBoard)

This appendix provides the complete set of supervision windows developed in ThingsBoard. Each state of the dashboard is illustrated with its dedicated snapshot and a brief description.

A.1 Home Dashboard

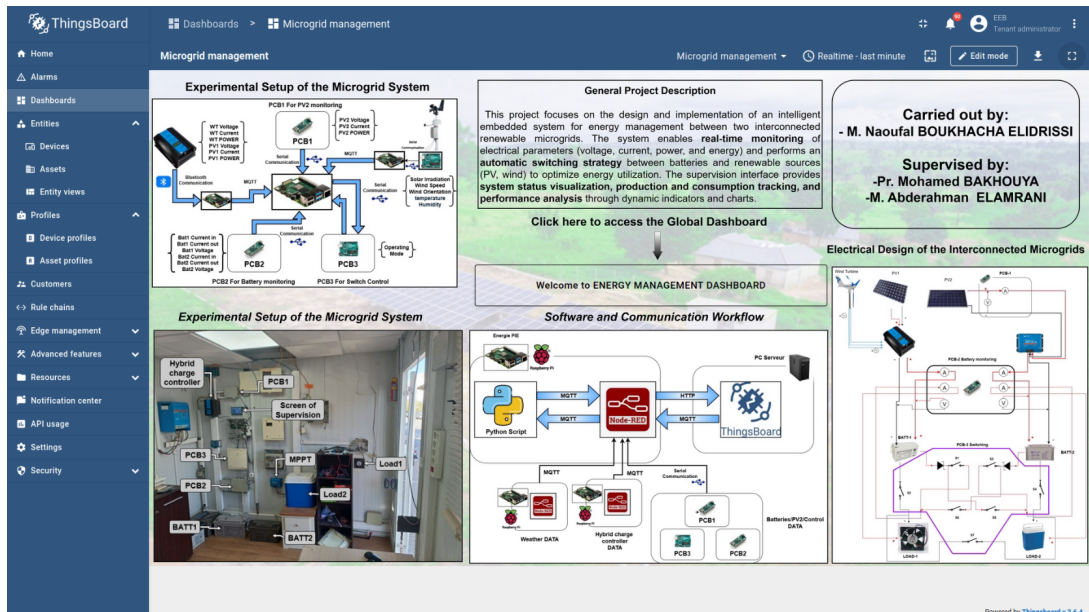


Figure A.1: Home dashboard: general description of the project and navigation button to access other states.

A.2 Global Dashboard

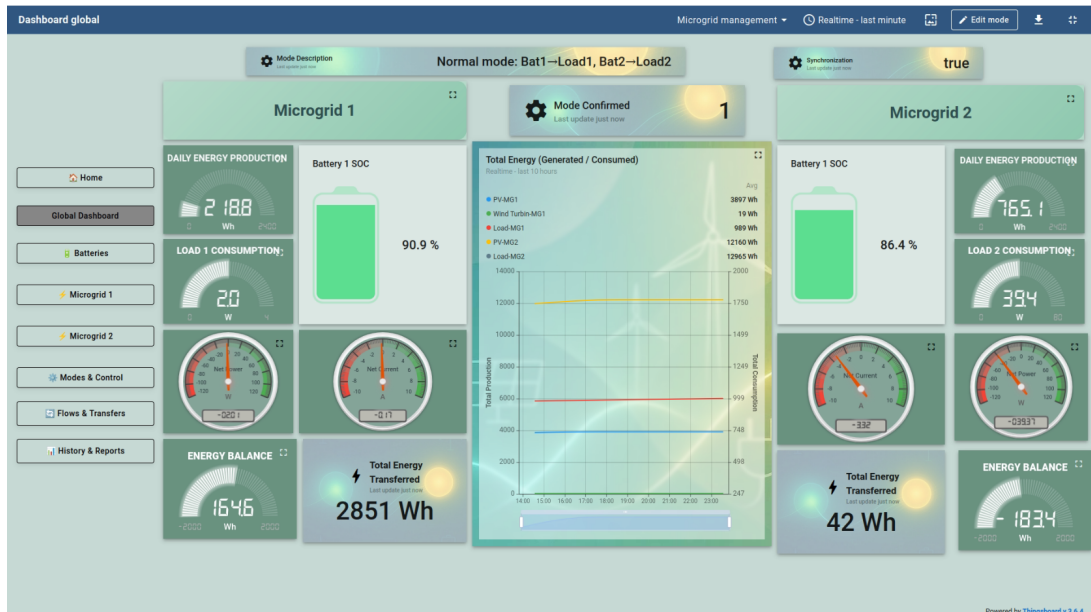


Figure A.2: Global dashboard: overview of both microgrids, SOC of Bat1 and Bat2, daily energy production, load consumptions, and total balance.

A.3 Battery Dashboard



Figure A.3: Battery dashboard: SOC, voltage, net current, net power, cumulative energy, and charge/discharge states.

A.4 Microgrid 1 Dashboard



Figure A.4: Microgrid 1 dashboard: PV1 production, wind turbine generation, load 1 consumption, and cumulative energies with daily balance.

A.5 Microgrid 2 Dashboard

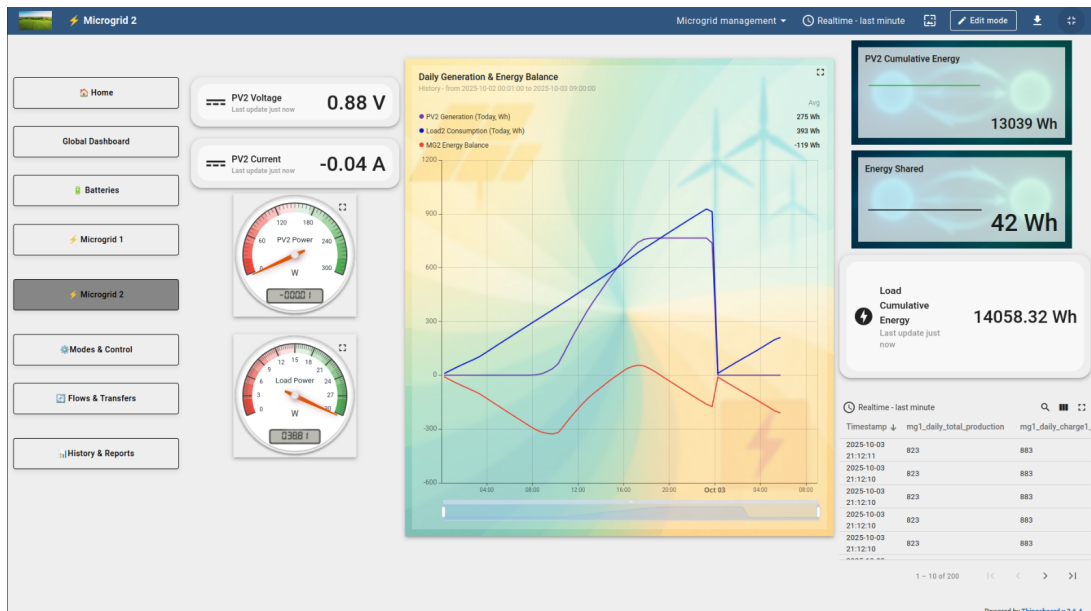


Figure A.5: Microgrid 2 dashboard: PV2 production, load 2 consumption, shared energy with MG1, and daily balance.

A.6 Modes and Control Dashboard



Figure A.6: Modes & control dashboard: recommended mode, confirmed mode, auto/-manual selector, and manual control buttons.

A.7 Flow and Transfer Dashboard



Figure A.7: Flow and transfer dashboard: cumulative energy shared between microgrids and timeline of switching modes.

A.8 History Dashboard

The screenshot shows the 'Historique & Rapports' (History & Reports) dashboard in ThingsBoard. The left sidebar contains navigation links: Home, Global Dashboard, Batteries, Microgrid 1, Microgrid 2, Modes & Control, Flows & Transfers, and History & Reports (selected). The main area displays a 'Timeseries table' for 'History' from 2025-10-02 13:05:00 to 2025-10-02 16:04:00. The table has 11 columns: Timestamp, mg1_pv1_voltage, mg1_pv1_current, mg1_pv1_power, mg1_wind_voltage, mg1_wind_current, mg1_wind_power, mg2_pv2_voltage, mg2_pv2_current, mg2_pv2_power, bat1_voltage, and bat1_current. The data shows alternating rows of solar and wind power generation. At the bottom, there is a pagination control showing 'Items per page: 30' and '1 - 30 of 50'.

Timestamp	mg1_pv1_voltage	mg1_pv1_current	mg1_pv1_power	mg1_wind_voltage	mg1_wind_current	mg1_wind_power	mg2_pv2_voltage	mg2_pv2_current	mg2_pv2_power	bat1_voltage	bat1_current
2025-10-02 16:03:59	13.1 V	1.4 A	18.3 W	0.0 V	0.0 A	0.0 W	27.9 V	3.4 A	100.7 W	13 V	1.4 A
2025-10-02 16:03:59	13.1 V	1.4 A	18.3 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	105.6 W	13 V	1.4 A
2025-10-02 16:03:58	13.2 V	1.4 A	18.5 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	105.6 W	13 V	1.4 A
2025-10-02 16:03:58	13.2 V	1.4 A	18.5 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	106.0 W	13 V	1.4 A
2025-10-02 16:03:58	13.2 V	1.4 A	18.5 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	106.0 W	13 V	1.4 A
2025-10-02 16:03:57	13.2 V	1.4 A	18.5 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	105.8 W	13 V	1.4 A
2025-10-02 16:03:57	13.2 V	1.4 A	18.5 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	105.8 W	13 V	1.4 A
2025-10-02 16:03:57	13.2 V	1.4 A	18.5 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	106.0 W	13 V	1.4 A
2025-10-02 16:03:56	13.2 V	1.4 A	18.5 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	106.0 W	13 V	1.4 A
2025-10-02 16:03:56	13.2 V	1.4 A	18.5 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	106.0 W	13 V	1.4 A
2025-10-02 16:03:55	13.1 V	1.4 A	18.3 W	0.0 V	0.0 A	0.0 W	29.4 V	3.6 A	106.0 W	13 V	1.4 A
2025-10-02 16:03:55	13.1 V	1.4 A	18.3 W	0.0 V	0.0 A	0.0 W	27.9 V	3.4 A	100.7 W	13 V	1.4 A
2025-10-02 16:03:55	13.1 V	1.4 A	18.3 W	0.0 V	0.0 A	0.0 W	27.9 V	3.4 A	100.7 W	13 V	1.4 A

Figure A.8: History dashboard: table of all parameters with selectable time range.

Appendix B

Validation Results

B.1 PV Validation

B.1.1 PV₁ Energy Validation

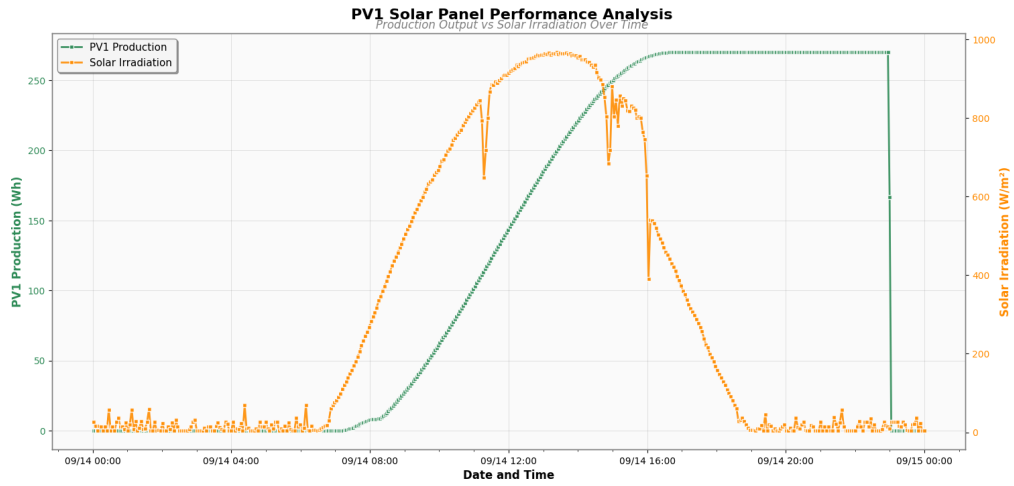


Figure B.1: Validation of cumulative energy production of PV₁ with respect to solar irradiation.

B.1.2 PV₂ Energy Validation

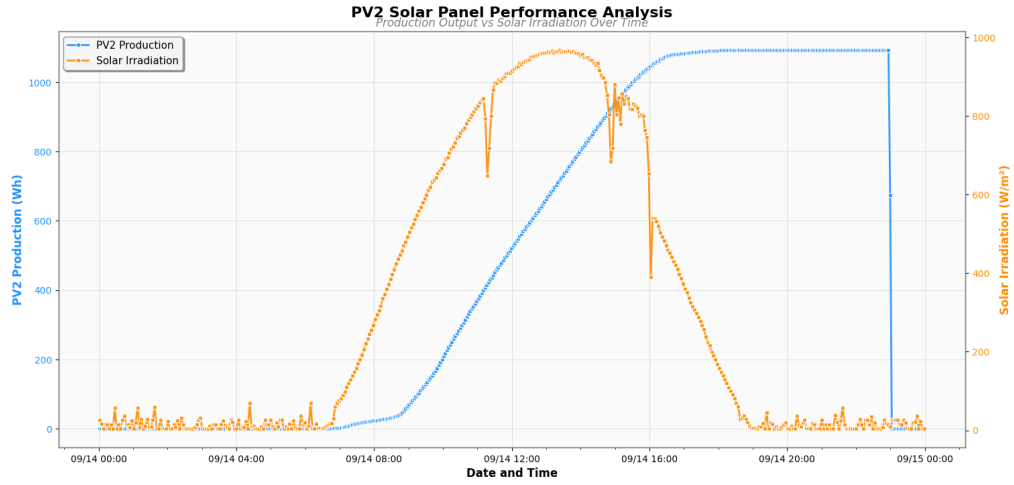


Figure B.2: Validation of cumulative energy production of PV₂ with respect to solar irradiation.

B.1.3 PV₁ Power Validation

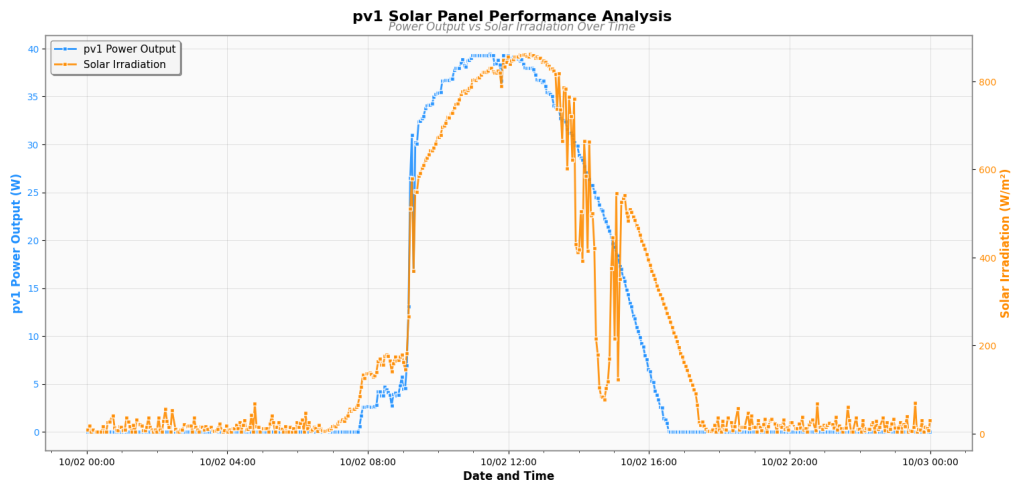


Figure B.3: Validation of instantaneous power of PV₁ with respect to solar irradiation.

B.1.4 PV₂ Power Validation

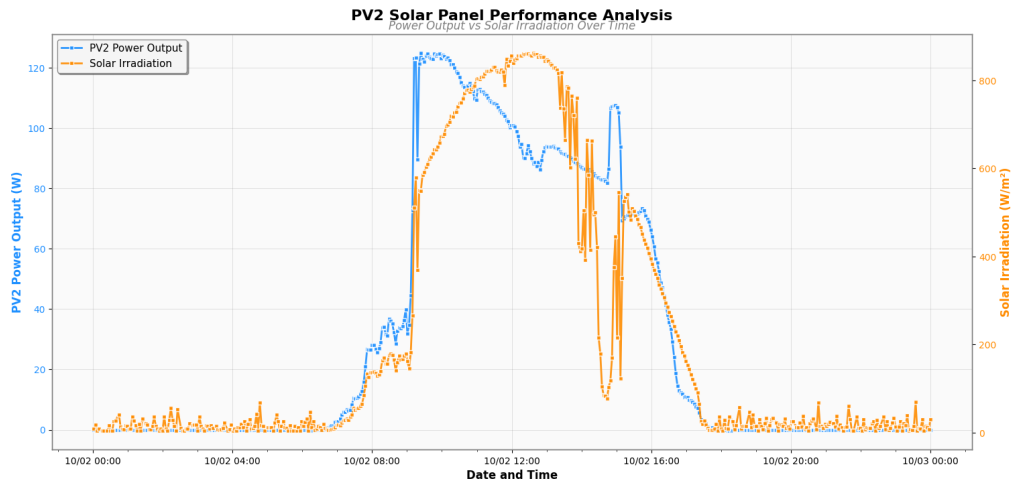


Figure B.4: Validation of instantaneous power of PV₂ with respect to solar irradiation.

B.2 Wind Validation

B.2.1 Wind Power Validation

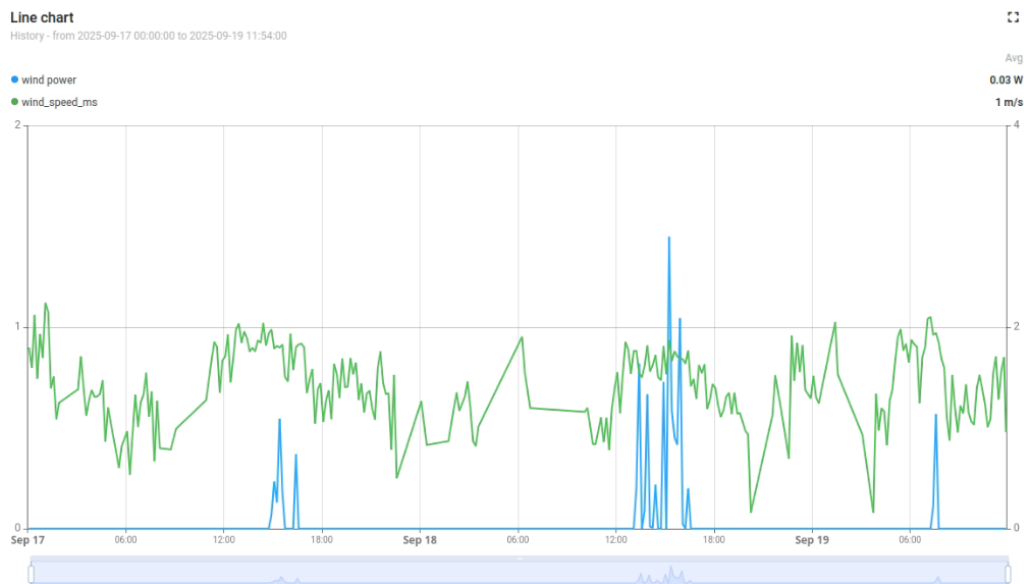
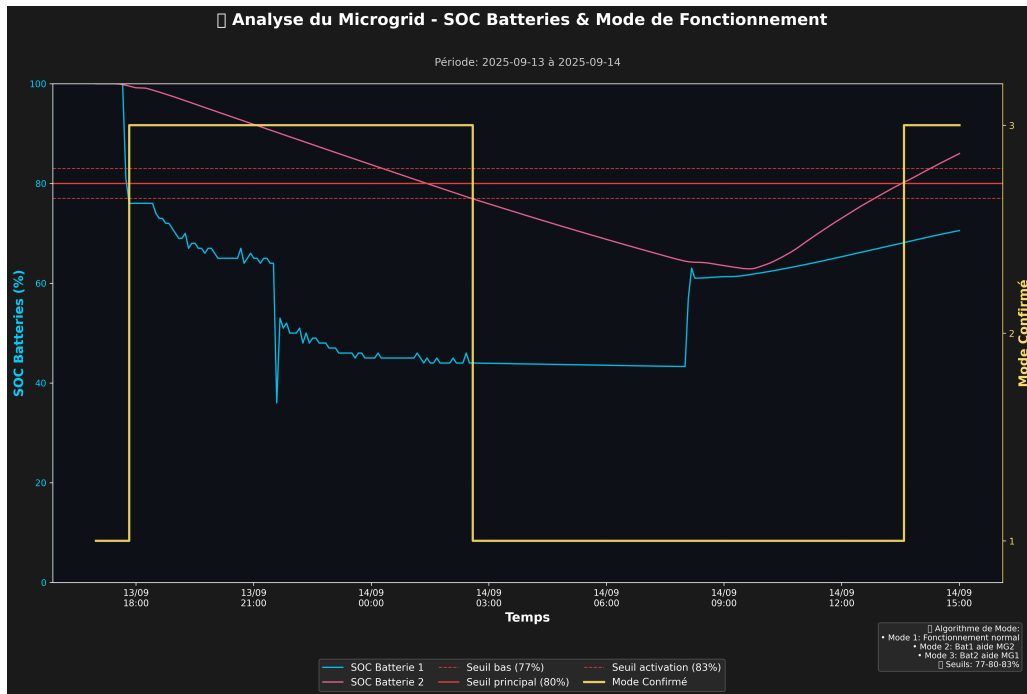


Figure B.5: Validation of wind turbine power production with respect to wind speed.

B.3 Switch-Based Control Validation

B.3.1 Battery and Load Management



Bibliography

- [1] 247 MESA. “Microgrid controllers: Functions and benefits.” Accessed: 2025-09-18. [Online]. Available: <https://247mesa.com/microgrid-controllers-functions-and-benefits/>
- [2] C. E. Reviews. “Pwm vs mppt solar charge controllers.” Accessed: 2025-09-18. [Online]. Available: <https://www.cleanenergyreviews.info/blog/mppt-solar-charge-controllers>
- [3] J. M. Guerrero, M. Chandorkar, T.-L. Lee, and P. C. Loh, “Microgrids: A review of technologies, key drivers, and outstanding issues,” *IEEE Transactions on Industrial Electronics*, vol. 58, no. 10, pp. 4789–4801, 2011.
- [4] R. H. Lasseter, “Smart distribution: Coupled microgrids,” *Proceedings of the IEEE*, vol. 99, no. 6, pp. 1074–1082, 2011.
- [5] Y. Liu, J. Li, W. Chou, and D. Xu, “Overview of different wind turbine concepts and their suitability for offshore wind farms,” *Renewable and Sustainable Energy Reviews*, vol. 64, pp. 479–491, 2017.
- [6] J. Zhang, Y. Wang, M. Li, and X. Chen, “A review on forecasting of renewable energy supply and demand using deep learning and big data,” *Applied Energy*, vol. 228, pp. 1649–1659, 2018.
- [7] H. Chen, T. Cong, W. Yang, Y. Tan, Y. Li, and Y. Ding, “Progress in electrical energy storage system: A critical review,” *Progress in Natural Science*, vol. 19, no. 3, pp. 291–312, 2009.
- [8] S. Vazquez, S. M. Lukic, E. Galvan, L. G. Franquelo, and J. M. Carrasco, “Energy storage systems for transport and grid applications,” *IEEE Transactions on Industrial Electronics*, vol. 57, no. 12, pp. 3881–3895, 2010.
- [9] L. He, X. Zhang, and C. Chen, “An optimal control strategy of battery energy storage system based on energy flow in microgrid,” *Proceedings of the CSEE*, vol. 31, no. 18, pp. 7–13, 2011.
- [10] D. T. Nguyen, W. Zhang, P. Liu, and H. Wang, “Optimal scheduling of battery energy storage system using model predictive control for microgrid applications,” *IEEE Transactions on Smart Grid*, vol. 10, no. 1, pp. 1069–1078, 2019.

- [11] D. Wu, Q.-C. Yang, F. Blaabjerg, and J. M. Guerrero, “Hierarchical control of microgrids—a review,” *Renewable and Sustainable Energy Reviews*, vol. 29, pp. 962–972, 2014.
- [12] H. Sun, W. Zhang, J. Li, and P. Wang, “Adaptive droop control for microgrids with multiple distributed generators,” *IEEE Transactions on Smart Grid*, vol. 8, no. 6, pp. 3027–3038, 2017.
- [13] W. Liu, H. Wang, M. Chen, and Y. Zhang, “Energy sharing management for microgrids with pv prosumers: A nash bargaining game approach,” *IEEE Transactions on Industrial Informatics*, vol. 14, no. 6, pp. 2154–2166, 2018.
- [14] C. Wang, X. Zhang, J. Li, and P. Liu, “Peer-to-peer energy sharing through a two-stage aggregated battery control in a microgrid,” *Applied Energy*, vol. 260, p. 114 246, 2020.
- [15] H. Farhangi, “The path of the smart grid,” *IEEE Power and Energy Magazine*, vol. 8, no. 1, pp. 18–28, 2010. DOI: [10.1109/MPE.2009.934876](https://doi.org/10.1109/MPE.2009.934876)
- [16] P. Almeida and J. Silva, “Embedded system for monitoring renewable energy microgrids,” *Renewable Energy*, vol. 139, pp. 1081–1091, 2019. DOI: [10.1016/j.renene.2019.02.094](https://doi.org/10.1016/j.renene.2019.02.094)
- [17] M. M. Rahman et al., “Low-cost data acquisition system for renewable energy applications,” in *Proceedings of the International Conference on Electrical Engineering and Renewable Energy*, IEEE, 2018, pp. 1–6.
- [18] R. Gupta and A. Kumar, “Embedded system for power monitoring in hybrid microgrids,” *International Journal of Renewable Energy Research*, vol. 10, no. 3, pp. 1250–1257, 2020.
- [19] S. Ahmad and M. Naeem, “Zigbee-based smart sensing and communication framework for microgrid monitoring,” *International Journal of Distributed Sensor Networks*, vol. 12, no. 7, pp. 1–9, 2016. DOI: [10.1177/1550147716653222](https://doi.org/10.1177/1550147716653222)
- [20] A. Singh and P. Sharma, “Wireless sensor network for real-time monitoring of renewable energy systems,” *Energy Reports*, vol. 7, pp. 2345–2354, 2021. DOI: [10.1016/j.egyrs.2021.05.022](https://doi.org/10.1016/j.egyrs.2021.05.022)
- [21] R. Gupta and A. Kumar, “Raspberry pi based energy management system for microgrids,” in *International Conference on Smart Energy Systems*, IEEE, 2021, pp. 50–56.
- [22] M. Ali and K. Hussain, “Real-time monitoring and control of microgrids using raspberry pi and python,” *International Journal of Smart Grid*, vol. 4, no. 2, pp. 89–96, 2020.

- [23] J. M. Guerrero, M. Chandorkar, T.-L. Lee, and P. C. Loh, “Advanced control architectures for intelligent microgrids—part i: Decentralized and hierarchical control,” *IEEE Transactions on Industrial Electronics*, vol. 60, no. 4, pp. 1254–1262, 2013. DOI: [10.1109/TIE.2012.2194969](https://doi.org/10.1109/TIE.2012.2194969)
- [24] N. Pogaku, M. Prodanović, and T. C. Green, “Modeling, analysis and testing of autonomous operation of an inverter-based microgrid,” *IEEE Transactions on Power Electronics*, vol. 22, no. 2, pp. 613–625, 2007. DOI: [10.1109/TPEL.2006.890003](https://doi.org/10.1109/TPEL.2006.890003)
- [25] A. Albarakati, A. Alghamdi, F. Alharthi, S. Alzahrani, M. Alsharif, and S. Kim, “An overview of microgrid control strategies for sustainable energy systems,” *Energies*, vol. 15, no. 3, p. 1052, 2022. DOI: [10.3390/en15031052](https://doi.org/10.3390/en15031052)
- [26] K. P. Schneider, D. Ton, R. H. Lasseter, P. Piagi, E. Bassi, et al., “Microgrid protection and control,” in *Proceedings of the 2018 71st Annual Conference for Protective Relay Engineers (CPRE)*, IEEE, 2018, pp. 1–13. DOI: [10.1109/CPRE.2018.8349771](https://doi.org/10.1109/CPRE.2018.8349771)
- [27] C. Simpkin et al., “Efficient orchestration of node-red iot workflows using a decentralized execution environment,” *Journal of Systems Architecture*, 2020, See DOI / publisher page for full reference (workflow/Node-RED research).
- [28] C. Y. Chen et al., “Web-based internet of things on environmental and lighting monitoring using node-red,” *Journal of Ambient Intelligence and Humanized Computing*, 2024.
- [29] OASIS, *Mqtt version 3.1.1*, Standard specification, 2014. [Online]. Available: <http://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html>
- [30] HiveMQ. “The history of mqtt.” Accessed: 2025-09-20. [Online]. Available: <https://www.hivemq.com/mqtt/history/>
- [31] ThingsBoard Inc. “Thingsboard documentation,” Accessed: Sep. 20, 2025. [Online]. Available: <https://thingsboard.io/docs/>
- [32] S. Ahmed and R. Kumar, “Design and implementation of an automatic transfer switch for hybrid microgrids,” *ARPJN Journal of Engineering and Applied Sciences*, vol. 14, no. 18, pp. 3130–3137, 2019.
- [33] L. Wang and M. Zhao, “Robust control of automatic transfer switching in microgrids,” *IEEE Access*, vol. 13, pp. 45 122–45 133, 2025. DOI: [10.1109/ACCESS.2025.3456789](https://doi.org/10.1109/ACCESS.2025.3456789)
- [34] S. Panda and K. Pradhan, “Soc-based load balancing in microgrids using distributed control,” *IET Renewable Power Generation*, vol. 16, no. 10, pp. 1920–1932, 2022. DOI: [10.1049/rpg2.12345](https://doi.org/10.1049/rpg2.12345)

- [35] X. Luo and Y. Zhang, “State-of-charge driven control strategy for microgrid switching,” *Applied Energy*, vol. 352, p. 121987, 2024. DOI: [10.1016/j.apenergy.2023.121987](https://doi.org/10.1016/j.apenergy.2023.121987)
- [36] *Silent wind 12v datasheet*, https://www.silentwindgenerator.com/wp-content/uploads/2018/11/SW400_Technical-Data_EN.pdf, 2020.
- [37] *Silent wind 12v wind turbine image*, <https://www.silentwindgenerator.com/fr/avantages/>, Accessed: 2025-09-21, 2020.
- [38] *Sl50ce-18 solar panel datasheet*, <https://generatory-wiatrowe.pl/produkty/ogniwa-fotowoltaiczne/>, Accessed: 2025-09-21, 2021.
- [39] *Silent wind 12v hybrid charge controller image*, <https://www.silentwindgenerator.com/fr/avantages/>, Accessed: 2025-09-21, 2020.
- [40] OUTDO, *Outdo 12v/150ah gel battery datasheet*, https://energymarket.ma/storage/files/4Tuj5pIkX0xHP3F91L3h_OT150-12GEL.pdf, Accessed: 2025-09-21, 2022.
- [41] *Co2-controlled ventilation system image*, <https://eu.mouser.com/ProductDetail/Delta-Electronics/FFB0812XHXHC>, Accessed: 2025-09-21, 2023.
- [42] *As-6p30-265w polycrystalline solar panel*, Datasheet available at: <https://www.weamerisolar.eu/wp-content/uploads/2017/03/AS-6P30-Module-Specification.pdf>, Amerisolar, 2025.
- [43] *Bluesolar mppt 150/35 charge controller*, Datasheet available at: <https://www.victronenergy.com/upload/documents/Datasheet-BlueSolar-charge-controller-MPPT-150-35-&-150-45-EN-.pdf>, Victron Energy, 2025.
- [44] *Victron energy gel battery 12v/130ah*, Datasheet available at: <https://www.victronenergy.com/upload/documents/Datasheet-GEL-and-AGM-Batteries-FR.pdf>, Victron Energy, 2025.
- [45] Arduino, *Arduino nano board datasheet*, Accessed: 2025-09-23, 2024. [Online]. Available: <https://docs.arduino.cc/hardware/nano>